Organic Chemistry Foundations

Ingyu Bahng

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Contents

1 Radical Halogenation

1.1 Foundations

Core Concept 1.1: Radical Halogenation

Radical halogenation is a substitution reaction in which a hydrogen atom of an alkane is replaced by a halogen atom, most commonly chlorine or bromine. Unlike conventional organic reactions that proceed through ionic intermediates, radical halogenation is mediated by **free radicals**—transient, highly reactive species characterized by the presence of an unpaired valence electron.

In synthetic organic chemistry, selective atomic substitution within molecular frameworks is a fundamental objective. However, alkanes present a significant challenge due to their chemical **inertness**; comprising exclusively strong, nonpolar C–C and C–H bonds, alkanes lack reactive functional groups that serve as recognition sites for electrophilic or nucleophilic reagents. Radical halogenation addresses this limitation by converting chemically inert alkanes into functionalized **haloalkanes** (*alkyl halides*), which possess reactive carbon-halogen bonds amenable to subsequent transformations. Consequently, radical halogenation represents a critical entry point into the broader landscape of functional group interconversions.

Core Concept 1.2: Homolytic vs. Heterolytic Cleavage

To understand radical formation, we must first examine bond cleavage mechanisms. Consider the **homolytic cleavage** of a generic covalent bond in compound AB.

$$\mathrm{A\!:\!B} \longrightarrow \mathrm{A\cdot} + \mathrm{\cdot B}$$

Homolytic cleavage results in symmetric bond fission, wherein each fragment retains one electron from the original bonding pair, yielding two **radical** species. The enthalpy change (ΔH) for this process corresponds to the bond dissociation energy, typically expressed in units of $kcal\ mol^{-1}$ and denoted as DH°.

This mechanism is fundamentally distinct from **heterolytic cleavage**, in which the bonding electron pair is retained entirely by one fragment, generating a cation-anion pair:

$$\mathrm{A\!:\!B} \longrightarrow \mathrm{A^{+}} + \mathrm{:\!B^{-}}$$

Importantly, the energy requirement for homolytic cleavage (DH°) typically exceeds that of heterolytic processes in polar solvents, as the latter benefit from solvation stabilization of the resulting ionic species.

Core Concept 1.3: Hydrocarbon C–H Bond Types

Alkane functionalization requires **C–H bond cleavage**, and the bond dissociation energy (DH°) exhibits a systematic decrease across the following homologous series.

$$\mathrm{H_3C{-}H} \quad > \quad \mathrm{R_{prim}{-}H} \quad > \quad \mathrm{R_{sec}{-}H} \quad > \quad \mathrm{R_{tert}{-}H}$$

The designations R_{prim}, R_{sec}, and R_{tert} refer to primary, secondary, and tertiary carbon centers, which are bonded to one, two, and three additional carbon atoms, respectively. These structural classifications are illustrated in Figure 1.

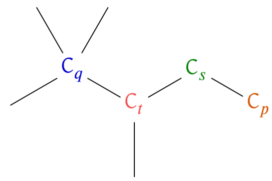


Figure 1: Primary (*orange*), secondary (*green*), tertiary (*red*), and quaternary (*blue*) carbons

The tertiary $R_{tert}-H$ bond exhibits the lowest bond dissociation energy and consequently generates the most thermodynamically stable radical upon cleavage. This stability trend—wherein radical stability increases with increasing alkyl substitution—arises from **hyperconjugation** (*Core Concept 1.4*).

Core Concept 1.4: Hyperconjugation

Hyperconjugation is a stereoelectronic stabilization mechanism wherein e^- density from a filled σ orbital (*typically a C-H or C–C bond*) delocalizes into an adjacent unfilled or partially filled p-orbital. In alkyl radicals, the bonding e^-s of σ bonds on substituent carbon atoms interact with the singly occupied p-orbital (*SOMO*) of the radical center, resulting in partial e^- delocalization that stabilizes the radical species.

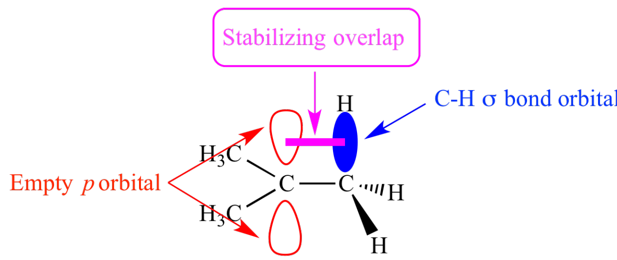


Figure 2: Hyperconjugation in alkane radicals

As illustrated in Figure 2, the magnitude of hyperconjugative stabilization is maximized in tertiary radicals ($\cdot R_{tert}$) due to the presence of a greater number of adjacent C–H or C–C σ bonds available for orbital overlap with the radical p-orbital. Each additional alkyl substituent provides additional filled σ orbitals capable of donating e^- density, thereby enhancing radical stability through cumulative hyperconjugative interactions.

1.2 Mechanism of Radical Halogenation

Core Concept 1.5: Initiation & Propagation

Having established the fundamental principles of alkane radical formation (*Subsection 1.1*), we now examine the mechanistic details of radical halogenation. The overall transformation, illustrated here using chlorine as the halogen, proceeds as follows.

$$H_3C-H + Cl-Cl \longrightarrow H_3C-Cl + HCl \qquad \Delta H = -25\ \text{kcal mol}^{-1}$$

Although the reaction is net exothermic ($\Delta H < 0$), it requires an initial activation energy to proceed. To elucidate the reaction pathway, we can decompose the overall transformation into the following elementary steps.

(1) **Initiation**

$$Cl_2 \longrightarrow 2\,Cl\cdot \qquad \Delta H = +58\ \text{kcal mol}^{-1}$$

Homolytic cleavage of the Cl–Cl bond can be induced through two primary mechanisms. **Thermal energy** elevates molecular vibrational energy to levels sufficient to overcome the bond dissociation energy, resulting in bond scission. Alternatively, **photochemical activation** via UV radiation promotes a bonding e^- to an antibonding molecular orbital ($\sigma \rightarrow \sigma^*$ *transition*), weakening the Cl–Cl bond and facilitating homolytic cleavage.

(2) **Propagation**

(a) $CH_4 + Cl\cdot \longrightarrow \cdot CH_3 + HCl \qquad \Delta H = +2\,\text{kcal mol}^{-1}$

The frontier molecular orbital interactions governing this hydrogen abstraction step are depicted in Figure 3.

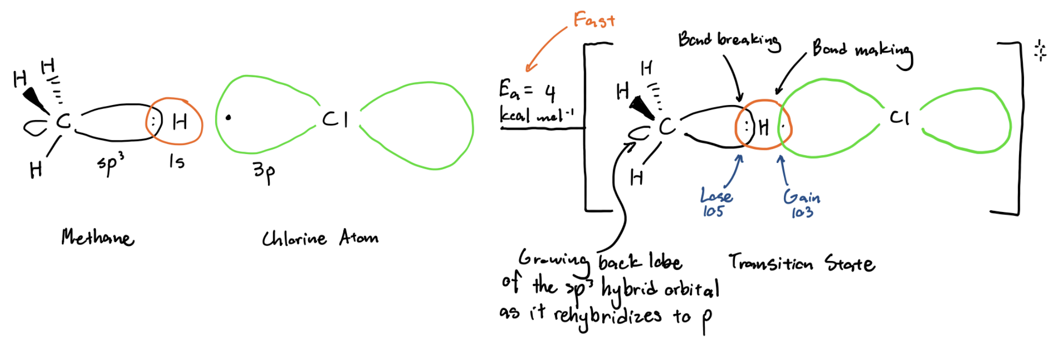


Figure 3: Orbital interaction of reagents in (2a)

This step yields a methyl radical ($H_3C\cdot$), characterized by a singly occupied 2p orbital and three sp^2-hybridized C–H bonds adopting trigonal planar geometry, along with HCl. In aqueous or polar environments, HCl, being a strong Brønsted acid, dissociates completely into solvated H^+ and Cl^- ions.

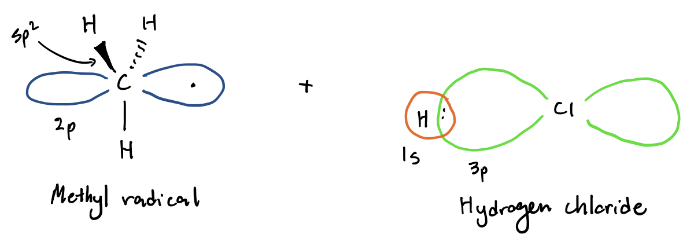


Figure 4: Products of (2a)

(b) $\cdot CH_3 + Cl_2 \longrightarrow CH_3Cl + Cl\cdot \qquad \Delta H = -27\,\text{kcal mol}^{-1}$

Summing the two propagation steps (2a) and (2b) regenerates the original stoichiometric equation.

$$CH_4 + Cl_2 \longrightarrow CH_3-Cl + HCl \qquad \Delta H = -25\,\text{kcal mol}^{-1}$$

Notably, the initiation step does not contribute to the net stoichiometry; it functions **solely to generate the initial radical species required to initiate the propagation cycle**. The chain-propagating nature of this mechanism means that a small initial population of Cl· radicals can facilitate the conversion of substantial quantities of starting materials, as each propagation cycle regenerates the chlorine radical. This autocatalytic regeneration enables each initiating radical to participate in numerous propagation cycles before termination, analogous to a catalytic turnover process.

The PE diagram for the propagation sequence (*Figure 5*) provides a quantitative representation of the energetic profile throughout the reaction coordinate.

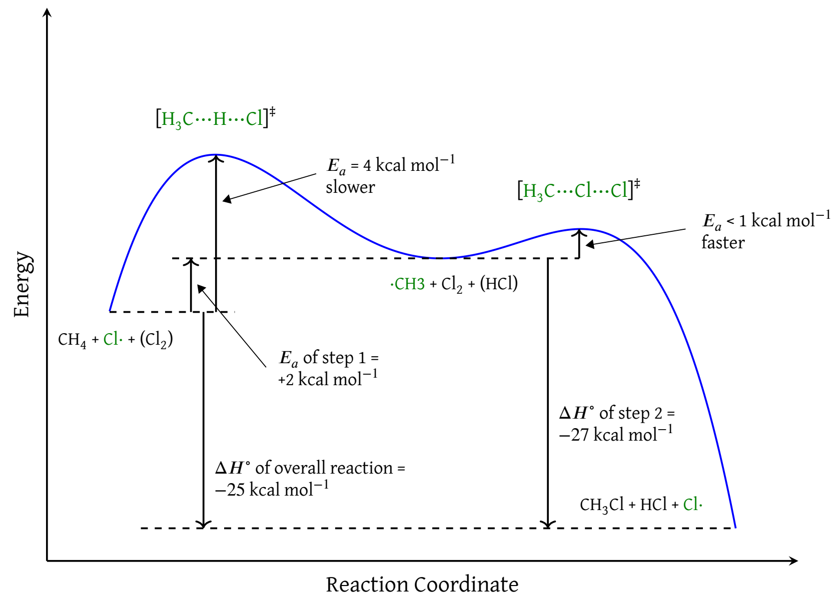


Figure 5: Potential energy diagram of halogenation propagation

Core Concept 1.6: Termination

Chain termination occurs through the following radical–radical coupling reactions.

(a) $2\,Cl\cdot \longrightarrow Cl_2$

This reaction represents the microscopic reverse of the initiation step, regenerating molecular chlorine through recombination of two chlorine radicals.

(b) $H_3C\cdot + Cl\cdot \longrightarrow CH_3Cl$

While this reaction formally produces the desired chloromethane product, it terminates the chain by consuming a propagating chlorine radical without regeneration, thereby halting further catalytic turnover.

(c) $H_3C\cdot + H_3C\cdot \longrightarrow H_3C-CH_3$

This dimerization reaction couples two methyl radicals to yield ethane, depleting the methyl radical pool required for propagation.

Throughout the reaction, radical intermediates such as $Cl\cdot$ and $H_3C\cdot$ exist at extremely low steady-state concentrations, reacting rapidly with abundant substrate (*CH_4 and Cl_2, respectively*) rather than persisting in solution. Consequently, termination reactions—requiring bimolecular radical–radical encounters—are negligible during early and intermediate stages. Termination becomes significant only when substrate depletion reduces propagation rates or when radical concentrations increase sufficiently in the closed system.

1.3 Halogen Reactivity and Selectivity

Core Concept 1.7: Reactivity of Other Halogens

Having examined chlorination in detail, we now extend our analysis to the full halogen series (*F, Br, and I*) to assess their respective reactivities.

DH° (+)				ΔH of H–X Formation (–)				ΔH of H_3C–X Formation (–)			
F_2	Cl_2	Br_2	I_2	HF	HCl	HBr	HI	CH_3F	CH_3Cl	CH_3Br	CH_3I
38	58	46	36	136	103	87	71	110	85	70	51

Table 1: Bond dissociation and formation enthalpies for halogens and halogen-containing compounds

- **Initiation energetics (left columns):** The bond dissociation energies for X_2 homolysis span a narrow range (36–58 kcal mol^{-1}). Since chlorination proceeds efficiently despite having the highest DH°, initiation is not rate-limiting under typical conditions.
- **Hydrogen halide formation (middle columns):** The exothermicity of H–X bond formation decreases systematically down the halogen group, with ΔH values ranging from -136 kcal mol^{-1} for HF to -71 kcal mol^{-1} for HI. HF formation is the most favorable (*most exothermic*) and HI formation is the least favorable (*least exothermic*) among the hydrogen halides.
- **Methyl halide formation (right columns):** An analogous trend is observed for H_3C–X bond formation, with fluorination ($\Delta H = -110$ kcal mol^{-1}) being substantially more exothermic than iodination ($\Delta H = -51$ kcal mol^{-1}).

Based on these thermodynamic data, we can predict that radical halogenation reactivity follows the order: $F > Cl > Br > I$, with fluorination being the most exothermic and iodination the least.

In practice, fluorination is **violently exothermic and uncontrollable**, often resulting in explosion. Conversely, **iodination does not proceed** under standard radical conditions due to its net endothermic character. **Chlorination and bromination occupy a synthetically useful middle ground**: chlorination is more reactive but less selective, while bromination is less reactive but more regioselective (*Core Concept 1.9*).

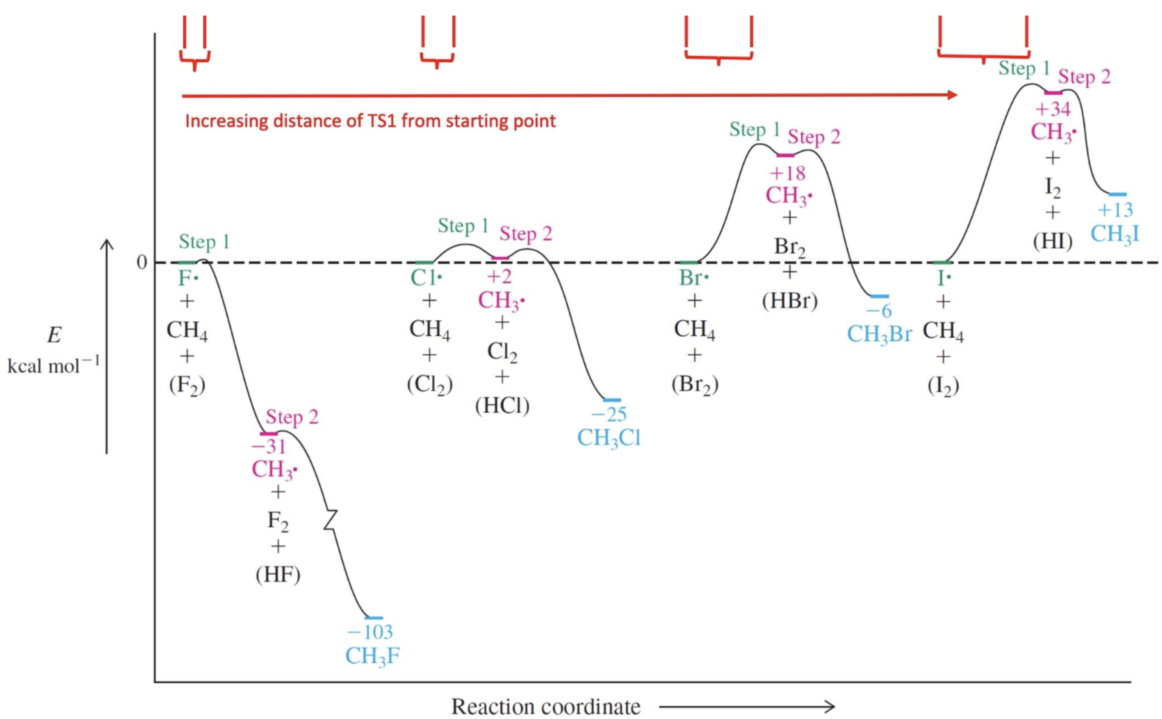


Figure 6: Comparison of Radical Halogenation PE Diagrams

Core Concept 1.8: Hammond Postulate

Inspection of Figure 6 reveals that the position of the first transition state (*TS1*) along the reaction coordinate shifts systematically as the endothermicity of the first propagation step increases. This observation is rationalized by the **Hammond Postulate**, a fundamental principle that relates transition state structure to reaction thermodynamics. Because transition states are transient, high-energy species that cannot be directly observed, the Hammond Postulate provides a framework for inferring their structures based on energetic considerations.

The Hammond Postulate states that for exothermic elementary steps, the transition state lies energetically and structurally closer to the reactants, resulting in an **early transition state** that resembles the starting materials more than the products.

Conversely, for endothermic elementary steps, the transition state lies closer to the products in both energy and structure, yielding a **late transition state** that more closely resembles the product species.

To illustrate this principle, consider the contrasting cases of fluorination and iodination, shown in magnified detail in Figure 7.

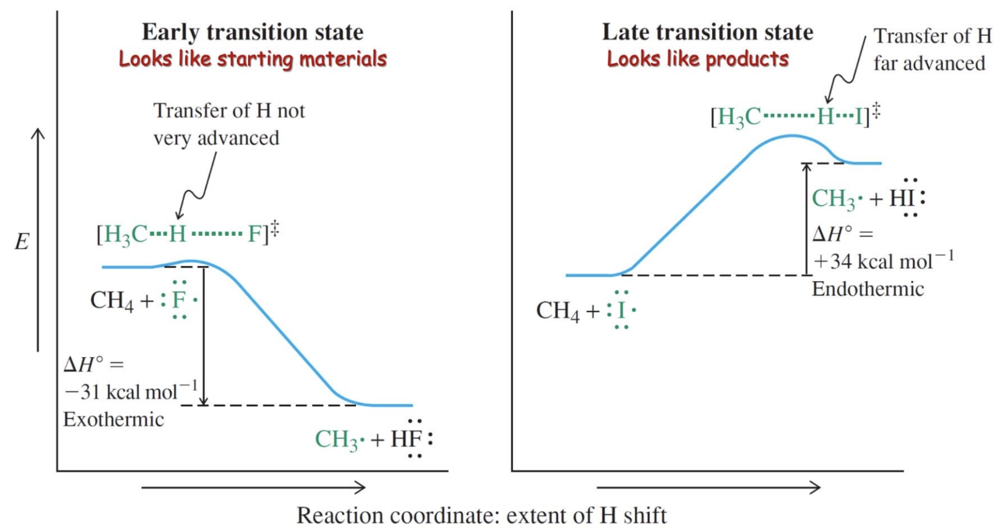


Figure 7: Zoomed Step 1 PE Diagrams for F and I

In the fluorination TS, the transferring hydrogen atom remains significantly closer to the methyl carbon than to the fluorine atom, reflecting the early, reactant-like character of this highly exothermic step. In contrast, the iodination TS exhibits the hydrogen atom positioned much closer to iodine, indicative of a late, product-like TS consistent with the endothermic nature of this step.

This structural distinction correlates directly with E_a differences. Fluorination exhibits a low E_a, requiring minimal C–H bond elongation to reach its TS; consequently, the C–H bond is only slightly weakened when the TS is achieved, and the hydrogen has not yet substantially transferred to fluorine. Iodination, conversely, requires a substantially higher activation energy, necessitating extensive C–H bond elongation and near-complete hydrogen transfer to iodine before the TS is reached. This reflects the weaker H–I bond strength relative to H–F, which requires greater geometric reorganization to compensate for reduced thermodynamic driving force.

Bromination, being mildly endothermic, similarly exhibits a late transition state, though less product-like than that of iodination.

Core Concept 1.9: Regioselectivity

The Hammond Postulate (*Core Concept 1.8*) also provides a mechanistic basis for understanding the differing **regioselectivities** exhibited by various halogens. To illustrate this, consider the competing pathways leading to primary versus tertiary isobutyl radicals:

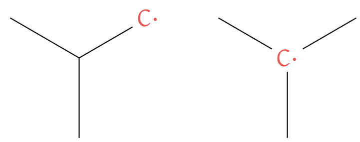


Figure 8: Isobutyl radical variations

Regioselectivity refers to the preferential abstraction of hydrogen from specific carbon centers (*methyl, primary, secondary, or tertiary*) based on the relative stabilities of the radical character developed in the transition state.

For chlorination, the **early transition state exhibits minimal radical character**, meaning that the differential stabilization between primary and tertiary radicals is not yet fully manifested at the transition state geometry. Consequently, chlorination displays poor regioselectivity, abstracting hydrogen atoms from primary, secondary, and tertiary positions at comparable rates.

Bromination, in contrast, proceeds through a **late transition state with substantial radical character**, wherein hyperconjugative stabilization of the developing radical center significantly influences the E_a. Because tertiary radicals benefit from greater hyperconjugative stabilization than primary radicals, the transition state leading to the tertiary radical is preferentially stabilized, resulting in highly regioselective hydrogen abstraction from tertiary positions.

Experimental selectivity data support this mechanistic interpretation, as shown in Table 2.

C–H Bond	F (25°C, gas)	Cl (25°C, gas)	Br (150°C, gas)
CH_3-H	0.5	0.004	0.002
RCH_2-H	1	1	1
R_2CH-H	1.2	4	80
R_3C-H	1.4	5	1700

Table 2: Relative reactivities of C–H bonds toward halogen radicals (*normalized to primary C–H*)

2 Nucleophilic Substitution and Elimination

2.1 Nucleophilic Substitution

After generating a haloalkane compound via **radical halogenation** (*Section 1*), the next step is to substitute that halogen with a different atom/molecule. One of the methods we can use to carry this out is nucleophilic substitution.

Core Concept 2.1: Nucleophilic Substitution

Nucleophilic substitution is a fundamental class of organic reactions where an e^- rich species, called a **nucleophile (Nu)**, attacks an electron-deficient atom called the **electrophile (E)**, displacing a specific group known as the **leaving group (L)**, which departs with the bonding electron pair. Halogens are common leaving groups due to their electronegativity and ability to stabilise the negative charge upon departure.

Nucleophilic substitution can be categorized into S_N2 and S_N1 .

Core Concept 2.2: S_N2

S_N2 reactions are **bimolecular**, meaning that both the leaving group and nucleophile are present in the transition state and making the reaction rate dependent on both [Nu] and [E-L] (Rate = $k[CH_3Cl][^{-}OH]$). The following is a chemical reaction of CH_3Cl undergoing S_N2 reaction with OH^-.

$$HO^- + CH_3{-}Cl \longrightarrow CH_3OH + Cl$$

Figure 9: S_N2 equation

Here, Cl is the leaving group and OH^- is the nucleophile. From a 3D point of view, we see the following outline interaction.

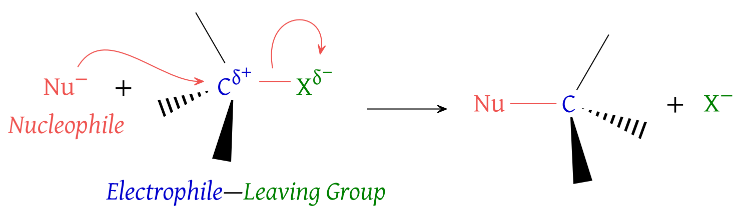


Figure 10: S_N2 mechanism reaction

It is important to note that S_N2 reactions occur in **one step**, where bond making and breaking occur simultaneously, meaning there are no intermediates—unlike the two-step propagation of radical halogenation (*Core Concept 1.5*). Consequently, the energy diagram consists of a single hill and is exothermic when the nucleophile is a more favorable attachment than the leaving group. *L is denoted as X in the figure below.*

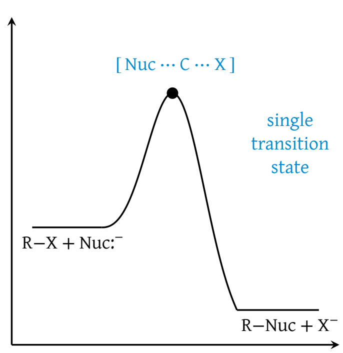


Figure 11: S_N2 energy diagram

A result of the reaction being one step is **stereospecificity**—there is *always* an inversion. A few key consequences of this are worth keeping in mind:

(a) Inversion does not always imply a shift between the R and S configuration (*Core Concept 4.25*)

(b) Two successive S_N2 reactions result in structural retention (double inversion)

(c) Performing S_N2 on only one chiral center of a molecule with two chiral centers yields a diastereoisomer (a non-mirror-image stereoisomer)

(d) Starting with a **racemate** (a 50/50 mix of R and S enantiomers) will still yield a racemate, as all configurations are inverted

All of these points follow intuitively from the structural consequences of the S_N2 mechanism, though they may not be immediately obvious.

Before exploring S_N1 reactions, it is worth establishing what defines a **"good" leaving group and nucleophile**, as this governs the outcome of both mechanisms.

Core Concept 2.3: Leaving Group Quality

There are two factors that determine whether a leaving group is good: **electronegativity** and **orbital size**. Regarding electronegativity, a good leaving group is also a weak base A^-—one that is stable with an extra e^- due to its high electronegativity. Regarding orbital size, a larger orbital size element forms relatively weaker bonds, as its molecular bonding orbitals occur at higher energy levels compared to smaller elements, making it more inclined to dissociate. Together, these two factors mean that better leaving groups appear moving left to right (→) and top to bottom (↓) across the periodic table.

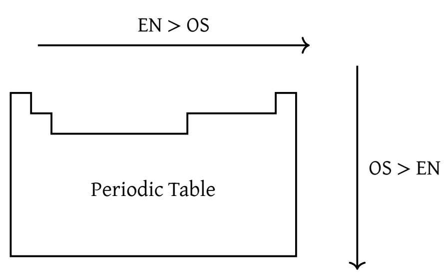


Figure 12: Leaving group ability within the periodic table

Comparing down the periodic table for halogens:

$$F^- < Cl^- < Br^- < I^-$$

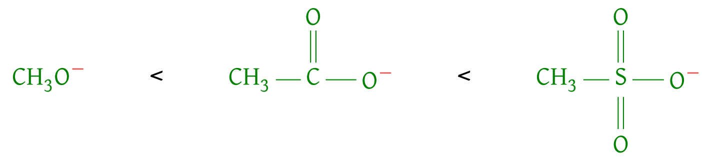
I^- is the best leaving group due to its large orbital size and correspondingly weak bond. Notably, I^- is such a great leaving group that it does not even participate in radical halogenation, making it impractical in that context. F^-, on the other hand, has the smallest orbital size, forming much stronger bonds through lower energy orbitals, making it the poorest leaving group of the halogens.

Comparing across the periodic table (*row 2*):

$$-CH_3 < -NH_2 < -OH < -F$$

Here, F^- is the best leaving group due to its greater electronegativity relative to O, N, and C, allowing it to better stabilise the extra e^- upon departure. In practice, however, F^- remains a poor leaving group relative to the broader periodic table, HO^- is exceedingly poor, and $NH_2{}^-$ and $CH_3{}^-$ do not leave at all.

For **polyatomic** leaving groups, **resonance** (π conjugation) plays an important additional role. The ability to delocalise e^-s across an adjacent p orbital system allows the leaving group to absorb excess charge more effectively—a larger delocalization system not only stabilises the extra charge but also weakens the leaving group's basicity, making it a better leaving group overall.



15.5 *poor* 4.7 *reasonable* -1.2 *excellent*

Figure 13: Polyatomic leaving groups; *corresponding numbers are pK_a values*

Core Concept 2.4: Nucleophile Quality

Nucleophilicity depends on four factors: **charge-basicity**, **basicity**, **solvation**, and **polarizability**.

(a) **Charge-basicity** refers to the greater tendency of an anion, relative to its neutral counterpart, to form a bond with a cation in order to delocalise its net negative charge.

$$OH^- > H_2O \qquad NH_2^- > NH_3 \qquad SO_4^{2-} > ROSO_3^-$$

(b) **Basicity** refers to how readily the nucleophile donates e^- density to the electrophile—analogous to how it would donate to H^+. As electronegativity decreases moving right to left across the periodic table, nucleophilicity increases accordingly.

$$H_3N > H_2O \qquad H_2N^- > HO^- \qquad HO^- > F^-$$

(c) **Solvation** refers to the ability of the surrounding solvent to impede the nucleophile's ability to attack the electrophile. In **polar protic solvents** (*such as water or ethanol, which contain O-H and N-H bonds*), strong hydrogen bonds effectively "handcuff" the nucleophile. Larger nucleophiles are less susceptible to this because their diffuse e^- clouds result in lower charge densities, weakening the solvent's grip.

$$F^- < Cl^- < Br^- < I^-$$

In **polar aprotic solvents** (*polar solvents lacking O-H and N-H bonds*), hydrogen bonding is absent and the nucleophile is largely unimpeded. The trend therefore reverses to reflect intrinsic basicity.

$$I^- < Br^- < Cl^- < F^-$$

Note that F^- remains a weak base in the grand scheme—it is simply less solvated than I^- in protic solvents, and does not compare to stronger bases such as OH^- or NH_2^-.

(d) **Polarizability** refers to the large, diffuse e^- clouds of heavier elements, which allow for more effective orbital overlap in the S_N2 transition state.

$$H_2O < H_2S < H_2Se \qquad \text{or} \qquad (CH_3)_3N < (CH_3)_3P$$

Nucleophilicity arises from a complex interplay of these factors. At its core, it is driven by **basicity**, which governs the horizontal periodic trend—atoms become less electronegative and more willing to donate e^- density moving left. The vertical trend, however, is governed by the solvent. In **polar protic solvents**, smaller nucleophiles such as F^- are heavily solvated, making larger, less-impeded ions like I^- the stronger attackers. In **polar aprotic solvents**, these hydrogen-bonding impediments are absent, allowing the intrinsically more basic smaller ions to reassert themselves as the stronger nucleophiles. **Polarizability** provides larger elements an additional advantage in S_N2 transition states, as their more diffuse e^- clouds can begin overlapping with the electrophilic carbon's orbital from a greater distance.

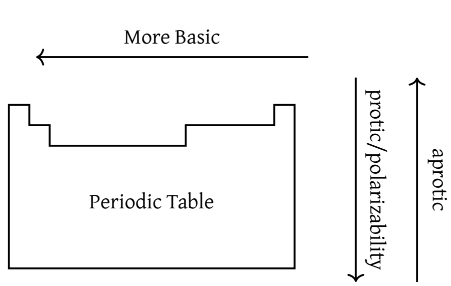


Figure 14: Nucleophilic ability within the periodic table

Core Concept 2.5: Steric Effects on Nucleophilic and Leaving Group Ability

Steric effects must be considered for both the nucleophile and leaving group. Steric effects arise from e^- repulsion between the nucleophile or leaving group and the three substituents bonded to the target electrophile.

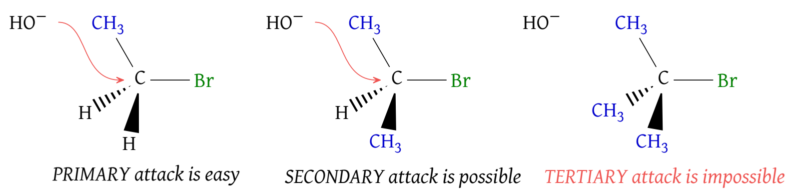


Figure 15: Steric effects in the context of S_N2 reactions

As a consequence, **leaving groups are better when larger**, as the repulsion assists their departure, while **nucleophiles are better when smaller**, as bulk impedes their approach. This steric hindrance is greatest when substituents are directly on the electrophilic carbon (α) and diminishes with distance (β, γ, δ, etc.).

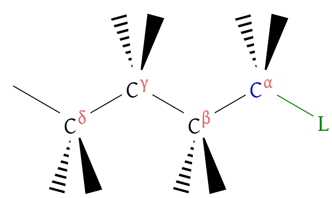


Figure 16: Carbon branching

Branch	Structure & Relative Rate			
	Hydrogens	Primary	Secondary	Tertiary
α	145	1	0.028	0
β	1	0.8	0.05	10^{-5}

Table 3: Rate of S_N2 reactions across different branching patterns

Core Concept 2.6: S_N1

S_N1 reactions are **unimolecular**, meaning the reaction rate is solely determined by [R – L].

$$\text{reaction rate} = k[\text{R–L}]$$

Unlike S_N2 , S_N1 proceeds through a series of distinct steps.

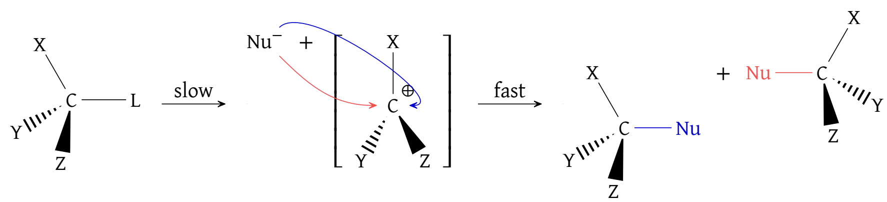


Figure 17: S_N1 mechanism

The first step—spontaneous dissociation of the leaving group from the electrophile—is the **rate limiting step**, as it is highly unfavourable, forming only trace amounts of carbocations at equilibrium. This makes it the bottleneck of the reaction, after which the nucleophile attacks either face of the sp^2-hybridised electrophile's perpendicular p orbitals.

Several important consequences follow from this mechanism.

(a) S_N1 nucleophilic attack is **not stereospecific**. A stereochemically pure reactant pool will yield a mixture of stereoisomers upon S_N1 . In practice, **slight inversion** is observed due to the transient physical blockade created by the lingering leaving group, which remains in proximity to the carbocation p orbital through ionic attraction.

(b) Because the rate is limited solely by carbocation formation, the **rate increases with a better leaving group**.

$$-Cl < -Br < -I < -O_3SR$$

(c) S_N1 **is accelerated by increasingly polar solvents**, particularly polar protic solvents. As the leaving group begins to depart, polar solvent molecules stabilise it through hydrogen bonding, effectively lowering the energy barrier for its departure.

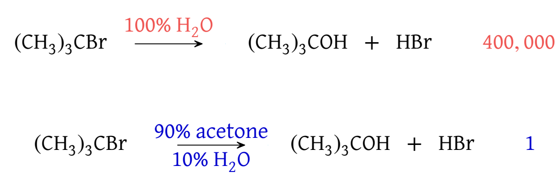


Figure 18: Relative S_N1 reaction rates in polar solvents

(d) **Product-determining steps occur after the bottleneck**. Once the carbocation intermediate forms, the dominant product is determined by nucleophilic competition—the strongest nucleophile in solution will attack preferentially. Consider, for instance, a mixture of NaCl and NaI in acetone.

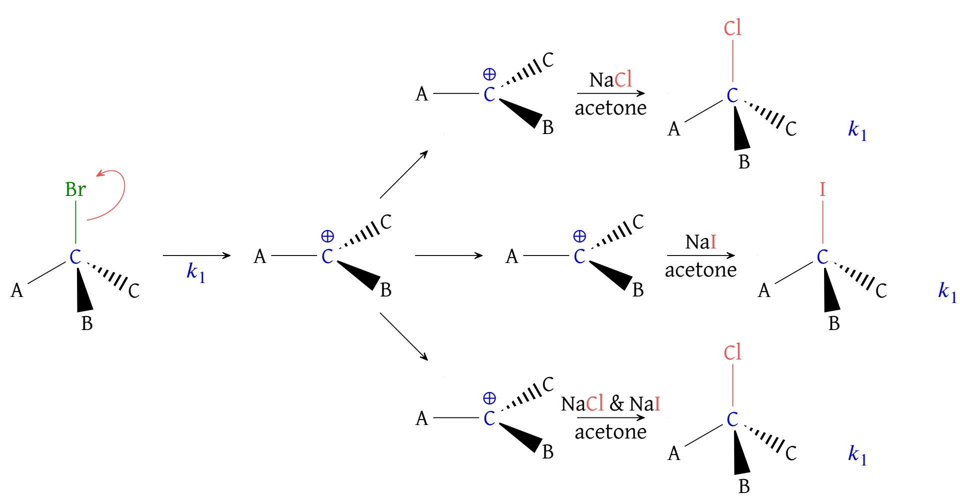


Figure 19: S_N1 nucleophile competition; *Cl is a better nucleophile than I, thus a mixture will result in the Cl product (bottom reaction)*

Here, the overall reaction rate is identical across all paths, as it depends solely on the departure of the leaving group (*regardless of the nucleophile identity*). Because acetone is a polar aprotic solvent, the Cl^- product will be favored under a mixture of Cl^- and I^- because Cl^- is a stronger nucleophile than I^- is.

Core Concept 2.7: Competition between S_N2 and S_N1

This raises an important question—if S_N1 and S_N2 ultimately achieve the same outcome, what drives a system to favour one over the other? Two factors determine this.

(a) α-branching disfavours S_N2 by sterically impeding nucleophilic approach, making it virtually impossible for a nucleophile to attack a tertiary electrophile in an S_N2 fashion.

(b) α-branched carbocations are stabilised by hyperconjugation with neighbouring molecular orbitals, lowering the energy of the rate-limiting step and making S_N1 viable.

$$\underbrace{H_3C^+ \text{ (methyl)} < H_3CH_2C^+ \text{ (primary)}}_{\text{energetically inaccessible in solution}} < \underbrace{(H_3C)_2HC^+ \text{ (secondary)} < (H_3C)_3C^+ \text{ (tertiary)}}_{\text{only cations feasible in solution}}$$

The table below summarises the conditions favouring S_N1 and S_N2 across different substrate classes. *Note that secondary R substrates do not inherently favour either mechanism—the outcome depends heavily on solvent and the nature of the nucleophile and leaving group.*

R	S_N1	S_N2
CH_3	**Not observed** (methyl cation too high energy)	**Frequent** (fast with good Nu/L)
Primary	**Not observed** (primary carbocation too high energy)	**Frequent** (fast with good Nu/L, slow when β branching present)
Secondary	**Relatively slow** (best with good L in polar protic solvents)	**Relatively slow** (best with high concentrations of good Nu in polar aprotic solvents)
Tertiary	**Frequent** (particularly fast in polar, protic solvents and good L)	**Extremely slow**

2.2 Elimination

Core Concept 2.8: Elimination

While substitution reactions involve replacing one group with another, **elimination** reactions involve **removal**—specifically, the loss of a leaving group and an adjacent hydrogen to form a new π bond, yielding an alkene. Like nucleophilic substitution, elimination reactions are classified into two types: E1 and E2.

Core Concept 2.9: Elimination 1 (E1)

Following carbocation formation in S_N1 , the nucleophile can alternatively act as a base (B^-), deprotonating a neighbouring alkyl group to yield a π bonded alkene over two distinct steps. *Note that this pathway requires a neighbouring alkyl group and is therefore not possible for methane.*

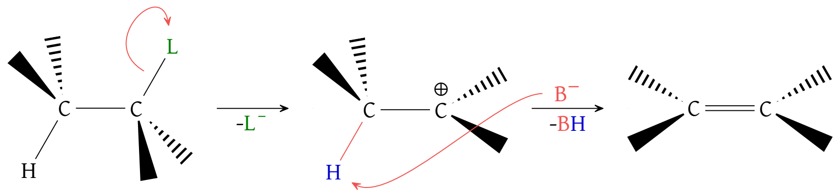


Figure 20: E1 mechanism

The rate-determining step is identical to that of S_N1, making E1 a first-order reaction whose rate depends solely on [R – L]. A closer look at the deprotonation step is provided below.

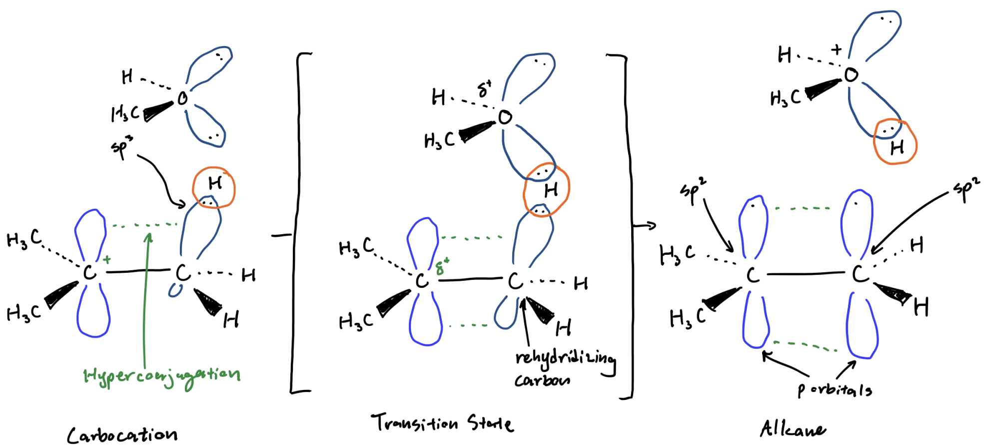


Figure 21: E1 deprotonation

Since deprotonation can occur at any neighbouring β C-H, E1 generally yields a mixture of alkenes across all possible pathways. Rotation around the resulting π bond gives rise to different stereoisomers.

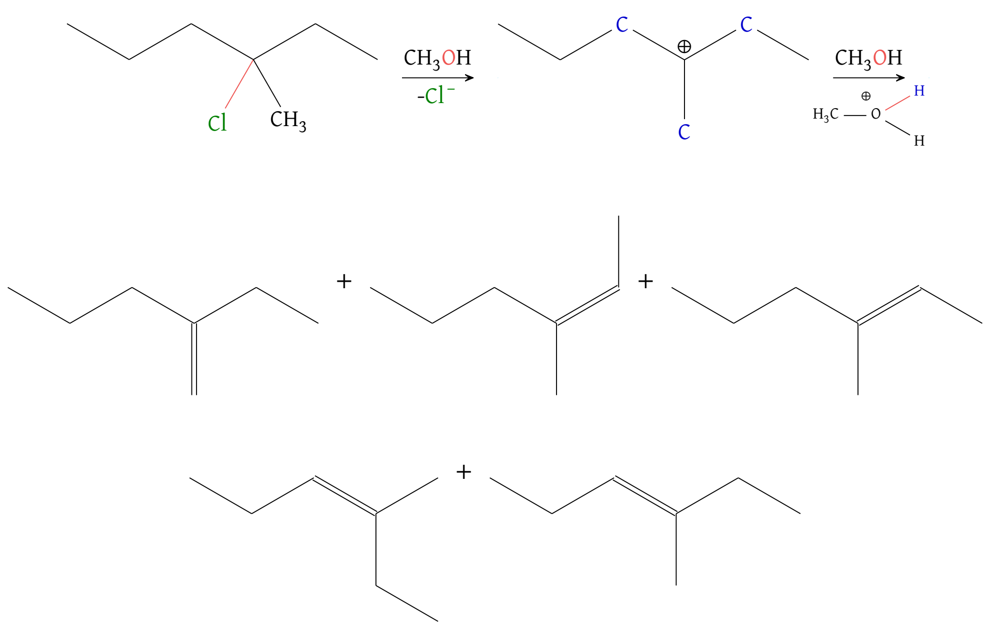


Figure 22: E1 regioisomeric distribution attributable to alternative deprotonation mechanisms

Crucially, S_N1 and E1 share the exact same rate-determining step—spontaneous departure of the leaving group to form a carbocation. Since the leaving group has already departed by the time the carbocation partitions between substitution and elimination, the S_N1 **to E1 product ratio is independent of L**.

The ratio between E1 and S_N1 products is difficult to predict precisely, but E1 is generally favoured under two conditions.

(a) **Higher temperature**, as elimination carries a positive entropy change: HRL + B $\longrightarrow$ R + L + BH.

(b) A **poor nucleophile**, as S_N1 requires the reagent to form a new C-Nu bond, demanding greater nucleophilic strength. E1 is less sensitive to this since the reagent acts purely as a base, abstracting a surface H^+ rather than integrating into the molecular framework.

But what about **strong bases**?

Core Concept 2.10: Elimination 2 (E2)

Strong bases attack R – L **directly** at the β C-H, constituting the E2 mechanism. E2 is faster than S_N1 /E1 and faster than S_N2 when sterically hindered by α or β branching, as it condenses the distinct steps of E1 into a single concerted process.

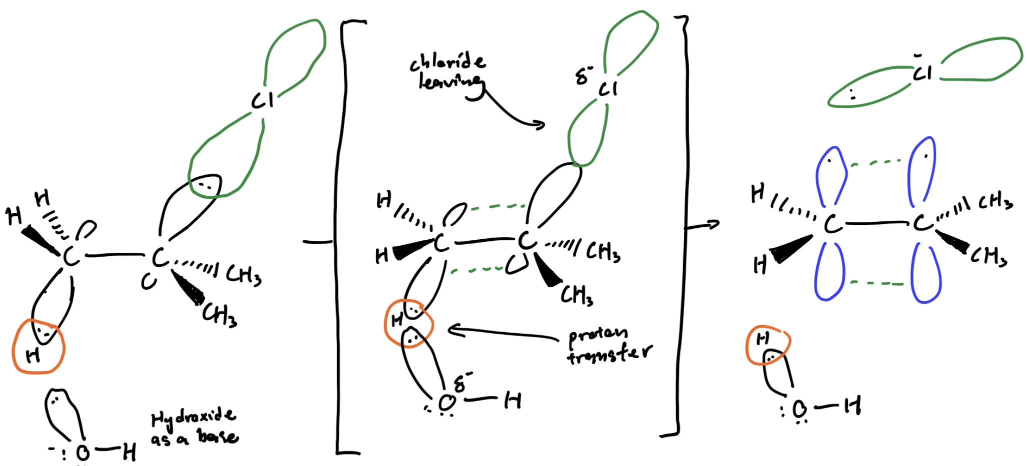


Figure 23: E2 transition state

Unlike E1, E2 is **second order** because the transition state involves both the substrate and the base.

$$Rate = k[\mathrm{R{-}L}][\mathrm{B^-}]$$

Consequently, a better leaving group and a stronger base both accelerate the reaction.

$$\mathrm{RCl} < \mathrm{RBr} < \mathrm{RI}$$

Additionally, electron-withdrawing substituents within the substrate accelerate E2 by increasing the acidity of the target β C-H through withdrawal of its e^- density.

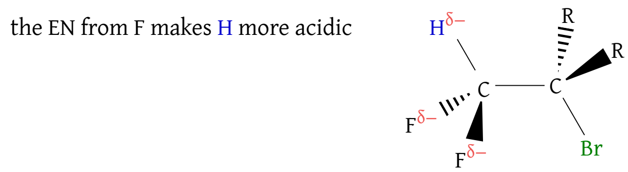


Figure 24: E2 hydrogen acidity amplification

E2 is also **stereospecific**—a given diastereomer of R – L yields only one stereoisomer of the alkene product. For instance, an RR/SS mixture yields exclusively one product, while an RS/SR mixture yields exclusively another.

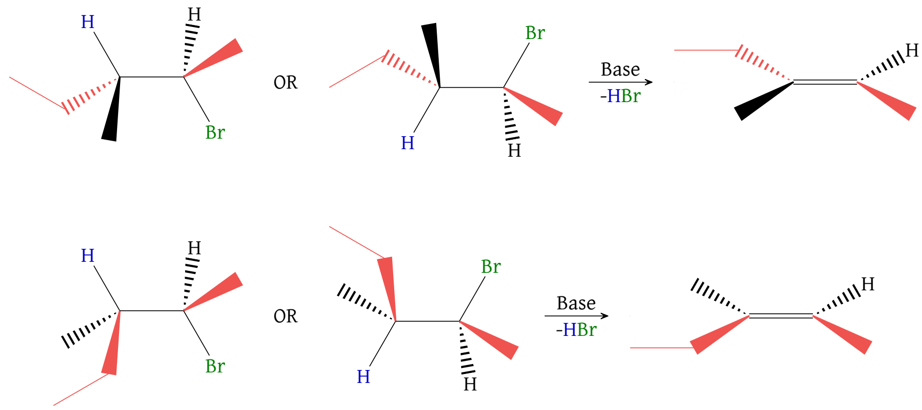


Figure 25: Stereospecific E2 outcomes; *RR (top left), SS (top right), SR (bottom left), RS (bottom right)*

This stereospecificity arises because H^+ and L depart simultaneously, requiring their initial positions to be coplanar to permit proper p orbital alignment in the forming π bond. This coplanarity can be achieved through either **anti**-periplanar (*trans*) or **syn**-periplanar (*cis*) geometry. When the relevant C-C bond can rotate freely, anti-periplanar geometry is strongly preferred as it minimises steric repulsion between the leaving group and the H – B transition complex.

In cases where free rotation is restricted—such as in cycloalkanes—syn-periplanar geometry must be adopted instead. This proceeds at a slower rate due to the increased steric hindrance between the leaving group and the H – B transition complex.

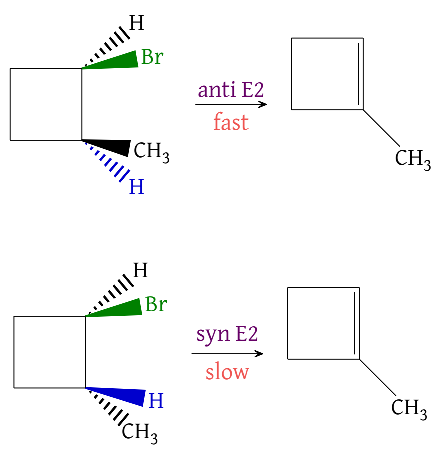


Figure 26: Anti and syn-periplaner E2 reaction rates

2.3 Competition

Having established the mechanics and characteristics of S_N2, S_N1, E1, and E2 reactions, it is important to recognise that all four pathways can be simultaneously possible across many overlapping conditions—and thus compete with one another in a given reaction environment. The table below organises these competing factors.

Type of haloalkane	Poor Nu (e.g. H_2O)	Weakly basic, good Nu	Strongly basic, unhindered Nu	Strongly basic, hindered Nu
Methyl	N/A	S_N2	S_N2	S_N2
Primary				
Unhindered	N/A	S_N2	S_N2	E2
Branched	N/A	S_N2	E2	E2
Secondary	Slow S_N1, E1	S_N2	E2	E2
Tertiary	S_N1, E1	S_N1	E2	E2

Core Concept 2.11: Factors of Competition between S_N2, S_N1, E1, and E2

Three core factors govern the competition between substitution and elimination.

(a) The base strength of the nucleophile

(b) Steric hindrance around the reacting carbon

(c) Steric hindrance around the nucleophile, particularly when it is a strong base

Primary substrates: Under poor nucleophile conditions, neither E1 nor S_N1 are viable due to insufficient hyperconjugation to sustain a carbocation, and neither S_N2 nor E2 proceed effectively as both depend on nucleophile strength. Moving from weakly basic to strongly hindered nucleophile conditions, the dominant factor becomes steric hindrance—whether along the haloalkane itself via β branching, or around the nucleophile through size or solvation in protic solvents. As hindrance increases, E2 is progressively favoured over S_N2.

Secondary substrates: Hyperconjugation is sufficient to support carbocation formation, making S_N1 and E1 viable under poor nucleophile conditions. E1 tends to dominate over S_N1 here, as abstracting a β H^+ is less demanding than integrating a nucleophile into the molecular framework. For stronger nucleophile conditions, the inherent steric hindrance of the secondary substrate generally drives the system toward E2, though S_N2 remains competitive when the nucleophile is weakly basic—insufficiently basic to abstract the β H^+ required by the E2 transition state.

Tertiary substrates: Steric hindrance completely shuts down S_N2, making S_N1 dominant under weakly basic, good nucleophile conditions. E2 dominates under strongly basic conditions, as it does for secondary substrates. Under poor nucleophile conditions, the S_N1 to E1 ratio becomes more balanced relative to secondary substrates—the greater stability of the tertiary carbocation lowers the activation energy for S_N1, allowing it to compete more effectively with E1.

Generally speaking, rather than memorizing each of these scenarios and their corresponding dominant mechanism, it's helpful to use S_N2 as a starting point and ask yourself the following questions.

(a) Is the nucleophile or haloalkane hindered? *If yes, the S_N2 reaction is likely not favored.*

(b) Is the nucleophile strongly basic? *If yes, E2. If no, S_N1/E1.*

However, keep in mind that there are other nuanced factors to consider such as absence of E1 preference for weakly basic, good nucleophile conditions.

3 Conformational Analysis

3.1 Rotamers

Core Concept 3.1: Rotamers

Rotamers, or rotational isomers, represent a specific class of conformational isomers characterized by distinct spatial arrangements arising exclusively from rotation about single (σ) bonds. Consistent with the fundamental principle established throughout Thermodynamics—that matter adopts configurations minimizing potential energy (PE)—molecular conformations similarly favor energetically favorable arrangements.

Core Concept 3.2: Rotamer Example (*Ethane & Propane*)

The conformational behavior of ethane can be effectively visualized through **Newman projections**, which reveal distinct rotameric states.

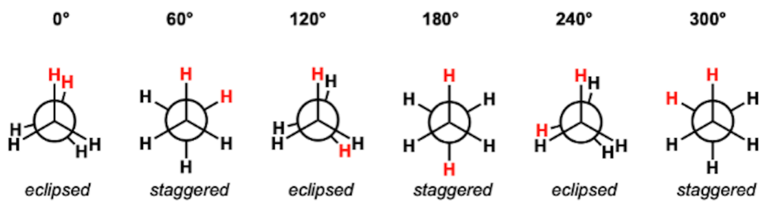


Figure 27: Newman projections of ethane

Rotation about the C–C σ bond generates two principal conformations: **eclipsed** and **staggered**. The staggered conformation represents the global energy minimum due to minimized e^-–e^- repulsions between C–H σ bonds. Conversely, the eclipsed conformation constitutes a local energy maximum, characterized by maximized e^-–e^- repulsions between adjacent C–H bonds. The PE profile for complete rotation about the C–C σ bond is depicted below.

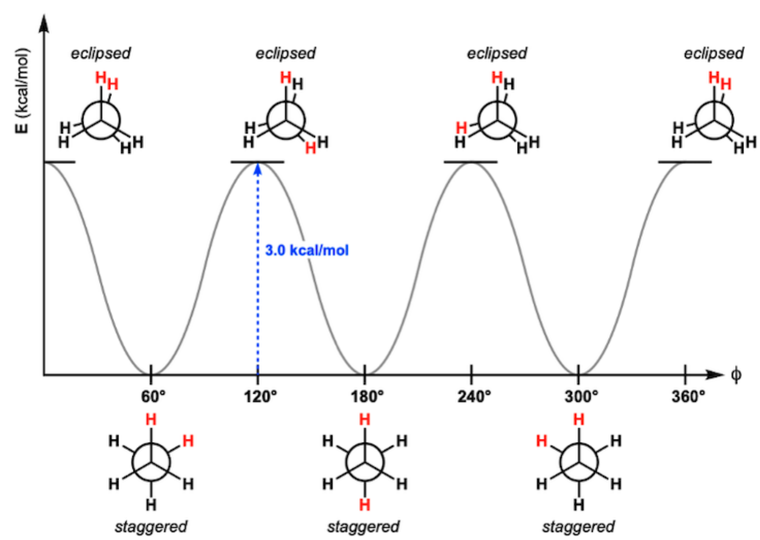


Figure 28: Potential energy diagram of ethane Newman projections

The barrier to rotation (3.0 kcal mol^{-1}) represents the activation energy (E_a) required for interconversion between staggered and eclipsed conformations. Notably, all staggered conformations of ethane are energetically degenerate, resulting in no net energy change upon rotation—the process is neither endothermic nor exothermic. This general pattern persists in propane, with a predictable modification.

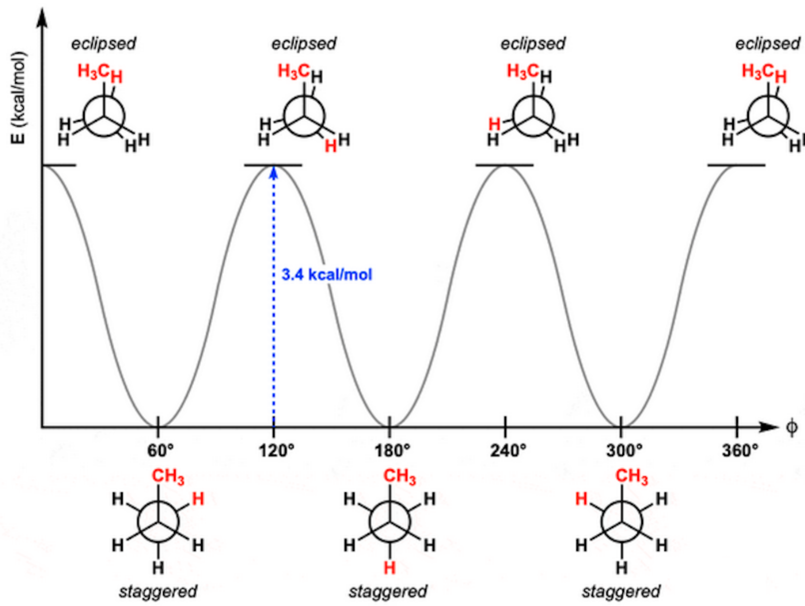


Figure 29: Potential energy diagram of propane Newman projections

The rotational barrier in propane increases to 3.4 kcal mol^{-1}, reflecting the enhanced e^- density of the methyl substituent relative to hydrogen in ethane. This increase in electron density amplifies e^-–e^- repulsive interactions, thereby elevating the energy barrier.

Core Concept 3.3: Rotamer Example (*Butane*)

Butane exhibits a more complex conformational landscape due to the presence of two methyl groups.

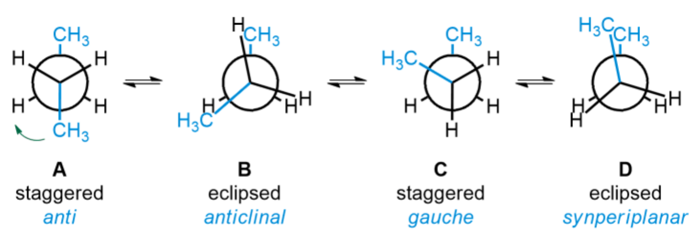


Figure 30: Newman projections of butane

Applying principles derived from propane's conformational analysis, the **anti** staggered conformation achieves maximum stability by positioning methyl groups at the greatest possible separation ($\theta = 180°$). The **gauche** staggered conformation ($\theta = 60°, 300°$) represents the next most stable arrangement. Among eclipsed conformations, the fully eclipsed state ($\theta = 0°$) exhibits minimum stability due to direct overlap of electron-dense methyl groups, while the half-eclipsed conformation ($\theta = 120°, 240°$) shows intermediate instability.

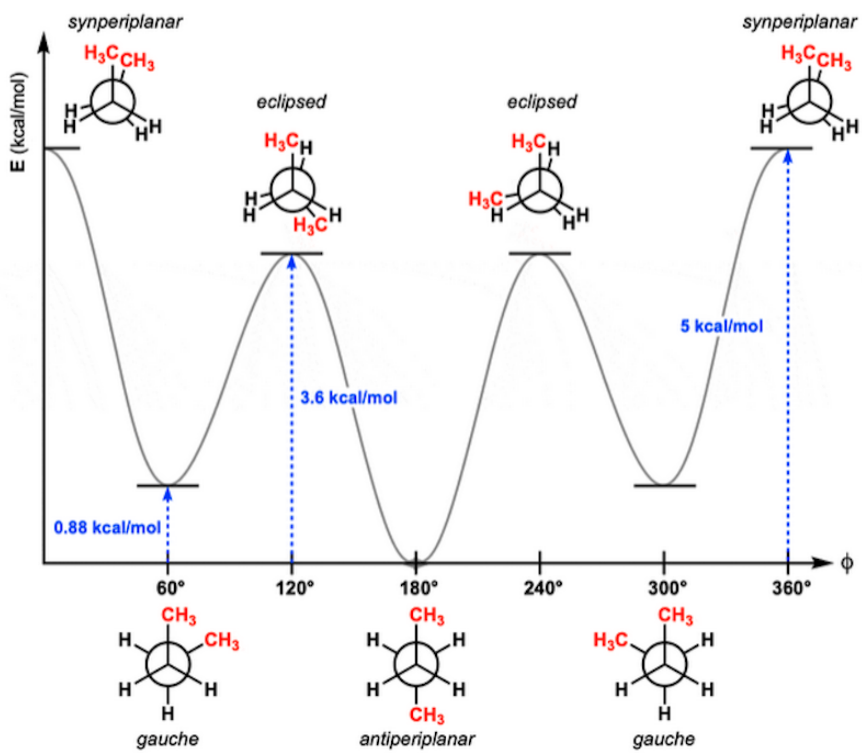


Figure 31: Potential energy diagram of butane Newman projections

Core Concept 3.4: Stereoisomeric Nomenclature

Stereoisomers are molecules with identical connectivity but differing three-dimensional spatial arrangements. This broad classification encompasses the following two categories.

(a) **Conformational isomers** arise from bond rotations that occur freely at room temperature, with rotamers serving as the archetypal example. These isomers interconvert rapidly without bond cleavage.

(b) **Configurational isomers**, in contrast, possess permanent stereochemical arrangements that can only be altered through bond-breaking and bond-forming processes. This category includes geometric isomers (*cis/trans*) and enantiomers (*R/S designations*).

3.2 Cycloalkane & Ring Strain

Core Concept 3.5: Cycloalkane Ring Strain

Cycloalkanes exhibit ring strain (*or elevated potential energy*) arising from the following three factors.

(a) **Bond angle strain**, particularly pronounced in smaller rings. Carbon atoms preferentially adopt sp^3 hybridization with ideal tetrahedral bond angles of 109.5°. However, geometric constraints imposed by ring closure force deviations from this optimal geometry.

For instance, cyclopropane exhibits internal bond angles of approximately 60°*, representing a* 49.5° *compression from the ideal value. This substantial angular distortion destabilizes the molecule by increasing orbital overlap inefficiency and raising the overall potential energy.*

(b) **Eclipsing strain** results from unfavorable torsional interactions between adjacent C–H bonds. As seen in Core Concept 3.2, eclipsed conformations maximize e^-–e^- repulsions between proximal σ bonds, elevating potential energy relative to staggered arrangements.

(c) **Transannular strain** is characterized by steric repulsions between atoms positioned across the ring interior. This strain component becomes significant in medium-sized rings (C_8–C_{11}) where the ring cavity is sufficiently small that non-bonded atoms on opposite sides of the ring approach at shorter distances.

These strain factors collectively drive cycloalkanes to adopt **non-planar conformations** that deviate from the intuitively expected flat ring geometry.

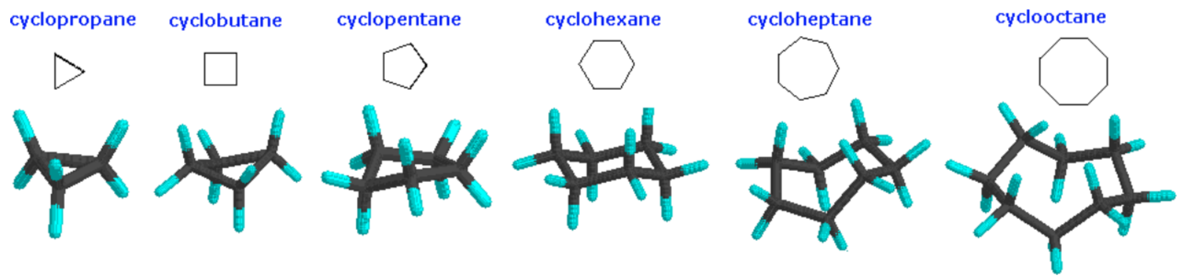


Figure 32: Non-planar cycloalkane conformations

Core Concept 3.6: Cyclohexane Conformational Analysis

Cyclohexane preferentially adopts the chair conformation (*Figure 33*) as its global energy minimum primarily due to the following reasons.

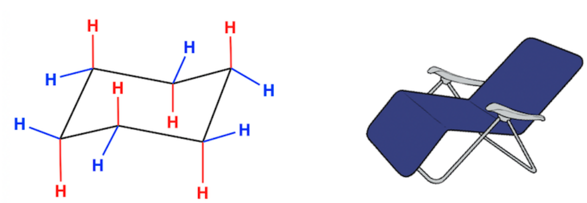


Figure 33: Cyclohexane chair conformation

(a) **Minimization of eclipsing strain:** The chair conformation's stability comes from the staggered orientation observed along all C–C bonds. Newman projection analysis reveals that each C–C bond maintains a **staggered** arrangement, thereby eliminating the eclipsing strain felt in planar conformations.

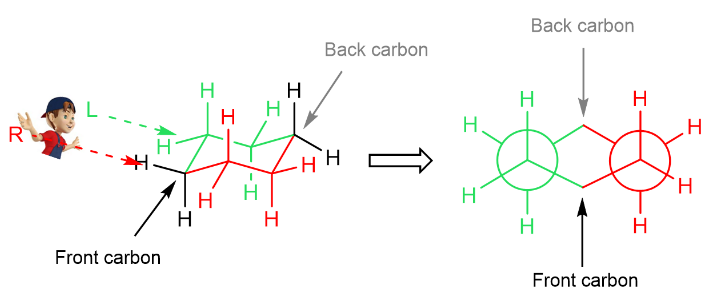


Figure 34: Newman projection analysis of cyclohexane

(b) **Minimization of transannular strain:** The chair conformation additionally minimizes steric interactions across the ring. In contrast, the **boat conformation** (*Figure 36*) suffers from both heightened transannular strain and increased eclipsing strain.

As a result, the boat conformation represents instead a **transition state** in cyclohexane ring inversion and consequently possesses negligible presence at equilibrium. On the other hand, the **twist (skew) boat conformation**—a geometrically relaxed intermediate—exhibits lower energy while retaining higher energy than the chair form (*Figure 36*).

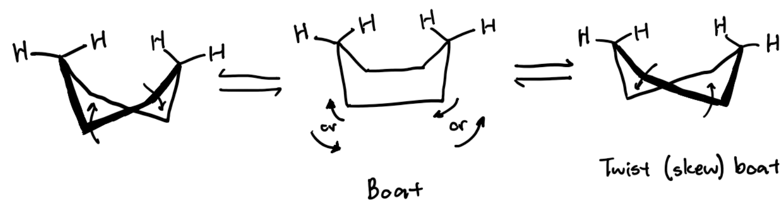


Figure 35: Cyclohexane twist (skew) boat conformation

Similarly to the boat conformation, the **half-chair conformation**—an additional high-energy transition state—arises from analogous strain considerations during the chair-to-chair interconversion pathway as shown in Figure 36.

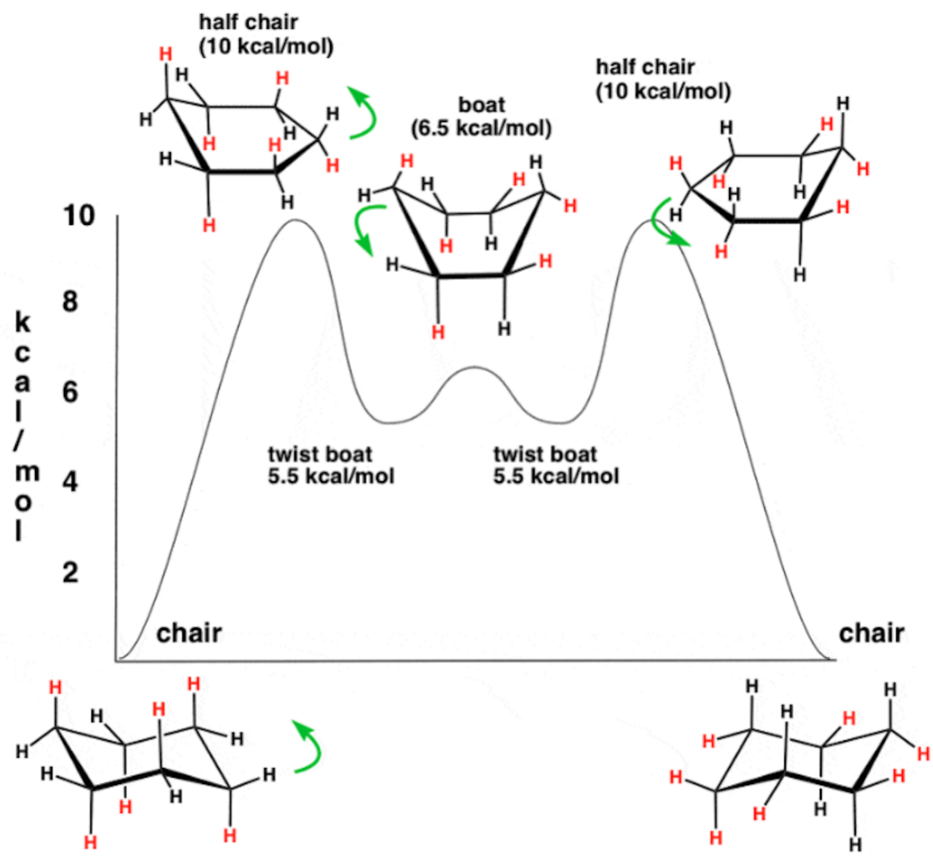


Figure 36: Cyclohexane conformation potential energy diagram

Core Concept 3.7: Monosubstituted Cyclohexanes

Whereas unsubstituted cyclohexane exhibits energetic degeneracy between chair conformations, monosubstituted derivatives (*e.g., methylcyclohexane*) demonstrate conformational preference driven by differential transannular strain.

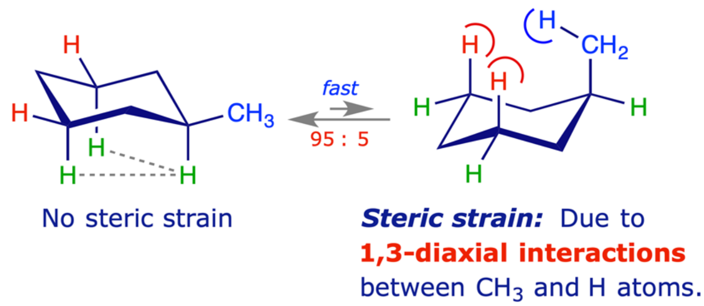


Figure 37: Methylcyclohexane chair conformations

Specifically, the **equatorial** methyl conformer predominates over the **axial** methyl conformer due to reduced 1,3-diaxial interactions as shown in Figure 37.

Core Concept 3.8: Axial vs Equatorial

Cyclohexane substituents occupy one of two geometrically distinct positions, each with characteristic spatial orientations and steric consequences.

(a) **Axial** bonds orient perpendicular to the approximate plane of the cyclohexane ring, extending parallel to the ring's pseudo-axis.

(b) **Equatorial** bonds, on the other hand, radiate outward approximately within the ring's equatorial plane, directed toward the ring periphery.

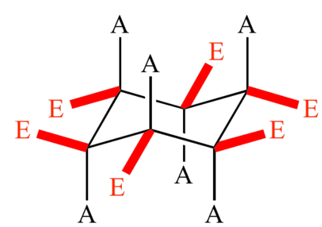


Figure 38: Axial (A) and equatorial (E) bonds within cyclohexane

4 Nomenclature

4.1 Alkanes

Core Concept 4.1: Alkanes

Alkanes are saturated hydrocarbons composed exclusively of carbon and hydrogen atoms connected by single covalent bonds. Nomenclature is determined by the number of carbon atoms in the chain: methane (CH_4) contains one carbon, ethane (C_2H_6) contains two, and so forth. All alkanes conform to the general molecular formula C_nH_{2n+2}, where n represents the number of carbon atoms.

Carbon Count	Alkane Name
1	Methane
2	Ethane
3	Propane
4	Butane
5	Pentane
6	Hexane
7	Heptane
8	Octane
9	Nonane
10	Decane
11	Undecane
12	Dodecane
13	Tridecane
14	Tetradecane

Table 4: Systematic nomenclature of alkanes by carbon number

4.1.1 n-Alkanes

Core Concept 4.2: Alkyl Substituents

An alkyl **substituent** is derived from its corresponding alkane by removal of a terminal hydrogen atom. Its general chemical formula is expressed as C_nH_{2n+1}, and its nomenclature follows that of alkanes, with the suffix "-ane" replaced by "-yl" (*e.g., ethyl, C_2H_5*).

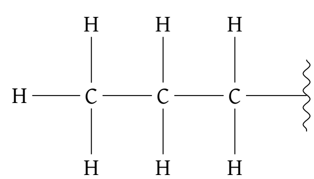
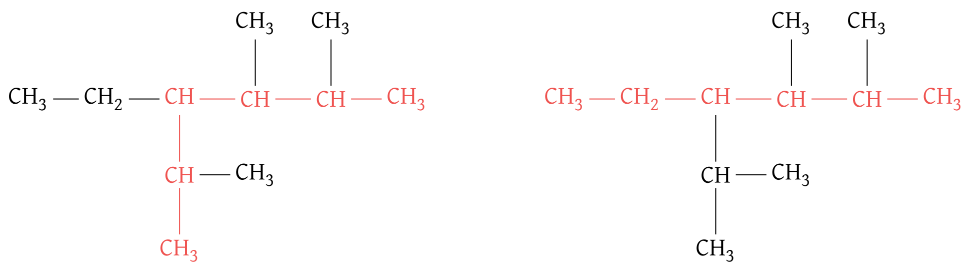


Figure 39: Propyl substituent

Core Concept 4.3: Foundational Alkane IUPAC System

The nomenclature of organic molecules adheres to a standardized system established by the **International Union of Pure and Applied Chemistry (IUPAC)**. The systematic naming of an organic molecule proceeds according to the following steps. *Nomenclature of subsequently covered organic compounds build upon the following process.*

(1) **Parent Chain Identification**: Identify the longest continuous alkane chain (*parent chain*) and assign its name based on the number of carbon atoms, as previously described in Table 4. When two chains of **equal length** exist, select the chain bearing the **greater number of substituents**.



Correct chain (*4 substituents*) Wrong chain (*3 substituents*)

Figure 40: Identifying the parent chain

In Figure 40, the longest chain contains 6 carbon atoms and is therefore designated **hexane**. Note that the left chain possesses more substituents than the right chain; consequently, the left chain is selected as the parent chain for nomenclature purposes.

(2) **Substituents**: Classify and name the substituents as either **alkyl** groups (*ethyl, propyl, butyl, etc.*) or **halo** groups (*bromo, fluoro, chloro, iodo*). Substituents will be denoted as *R* throughout this section.

(a) **Unbranched Substituents**: For an unbranched substituent R, apply the naming convention defined in step (2) directly.

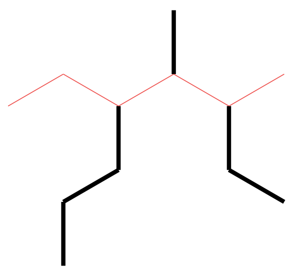


Figure 41: Unbranched substituents; propyl (*left*), methyl (*middle*), ethyl (*right*)

(b) **Branched Substituents**: For a branched substituent R, identify the longest carbon chain within R and systematically name any substituents attached to this chain.

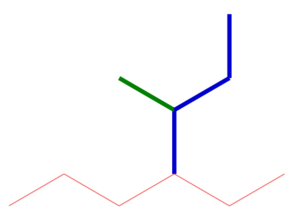


Figure 42: Methyl propyl branched substituent; methyl (*green*), propyl (*blue*)

(c) **Multiple Identical Substituents**: When multiple identical substituents are present, employ the prefixes di-, tri-, tetra-, etc. for unbranched R groups and bis-, tris-, tetrakis-, etc. for branched R groups.

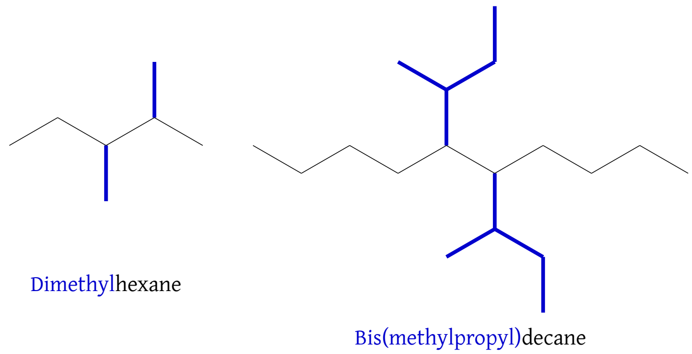


Figure 43: Multiple straight (*left*) and branched (*right*) chain substituents

(d) **Common Substituent Names**: While systematic nomenclature exists for all alkyl substituents, certain common names remain in widespread use and must be committed to memory.

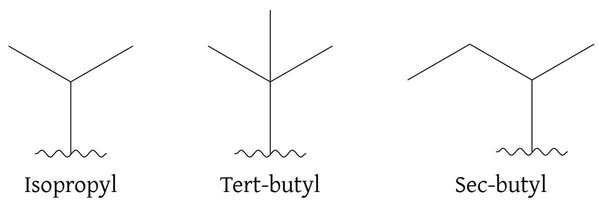


Figure 44: Common alkyl substituent names

(3) **Position Numbering**: Number the carbon atoms along the parent chain to indicate substituent positions. The numbering sequence initiates from the terminus **nearest to a substituent**. If both termini are equidistant from the first substituent, continue comparison until reaching the **first point of difference**.

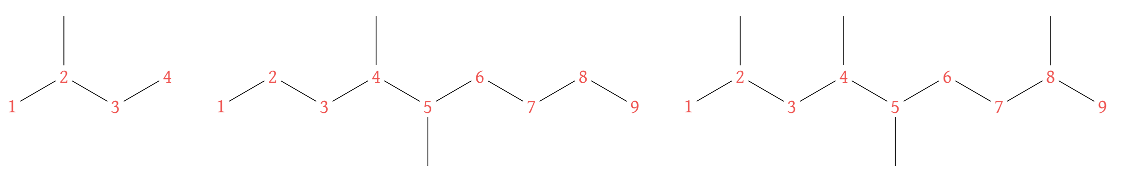


Figure 45: Numbering of parent chain in the order of earliest substituent presence (*left & middle*) and first point of difference (*right*)

If substituent **positions remain identical** from either direction, initiate numbering from the terminus where the first substituent's name appears **alphabetically earliest**.

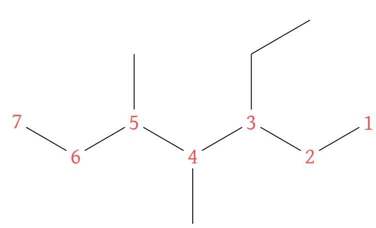


Figure 46: Alphabetical numbering of parent chain when substituent position is mirrored; *ethyl precedes methyl alphabetically, therefore numbering commences from the right terminus*

This numbering protocol applies equally to the parent chain within a branched substituent. However, numbering invariably begins at the carbon atom of attachment to the main chain.

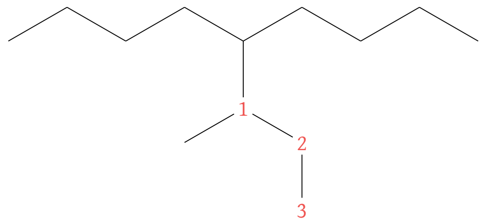
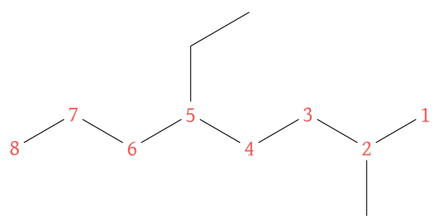


Figure 47: Branched chain numbering

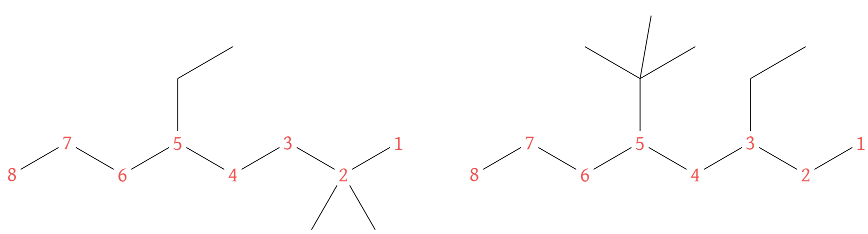
(4) **Alphabetical Substituent Ordering**: Having established both the identities and positions of substituents, the complete name of the alkane can now be constructed. Note that substituent names are ordered **alphabetically** rather than numerically. Positional designations are indicated by numerical prefixes corresponding to each substituent.



5-Ethyl-2-methyloctane

Figure 48: Alphabetical ordering of substituents

Note that multiplicative prefixes such as di-, tri-, and bis- are **disregarded** when alphabetizing substituents on the parent chain, but **are considered** when alphabetizing within complex branched substituent names.



5-Ethyl-2,2-dimethyloctane 5-(1,1-Dimethylethyl)-3-ethyloctane

Figure 49: Alphabetical counting of prefixes; *the d (blue) in the left compound is disregarded because it refers to parent chain substituents*

A comprehensive example illustrating these principles is presented below.

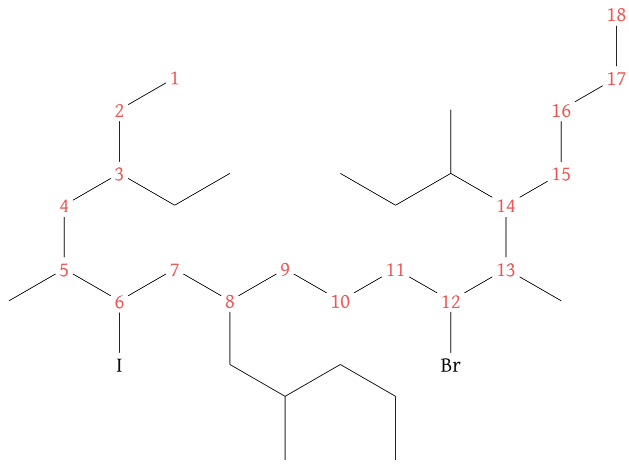
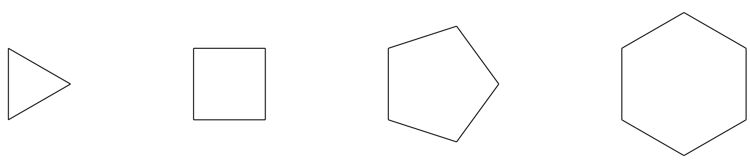


Figure 50: 12-bromo-3-ethyl-6-iodo-5,13-dimethyl-8-(2-methylpentyl)-14-(2-methylpropyl)octadecane

4.1.2 Cycloalkanes

Core Concept 4.4: Cycloalkanes

Cycloalkanes are cyclic alkane structures named systematically according to the number of carbon atoms in the ring, with the prefix ”cyclo-” preceding the parent alkane name.



Cyclopropane Cyclobutane Cyclopentane Cyclohexane

Figure 51: Cycloalkanes

Core Concept 4.5: Cycloalkane IUPAC System

The systematic nomenclature of cycloalkanes follow the procedure outlined below.

(1) **Parent Chain Identification**: Identify the body that possesses the greatest number of carbon atoms.

(a) **Simultaneous Presence of Cycloalkane and n-Alkane**: The parent chain is designated as the unit possessing the greater number of carbon atoms.

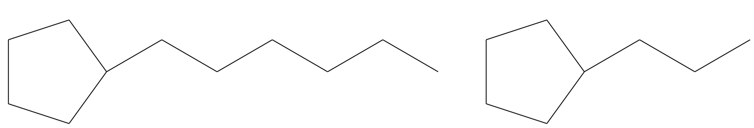


Figure 52: 1-cyclopentylhexane (*left; cycloalkane as substituent*) and propylcyclopentane (*right; chain alkane as substituent*)

Note that for propylcyclopentane in Figure 52, a locant for the propyl group is unnecessary, as only one substituent is present on the parent cycloalkane and carbon numbering would therefore be arbitrary.

(b) **Multiple Cycloalkane Units**: The smaller ring functions as a substituent to the larger ring.

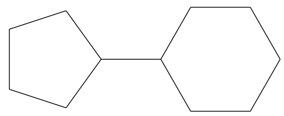
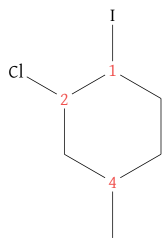
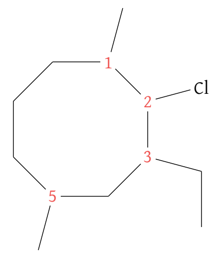


Figure 53: Cyclopentylcyclohexane

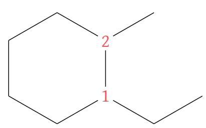
(2) **Substituents**: When multiple substituents are attached to a cycloalkane ring, carbon atoms are numbered to yield the **lowest possible digit sequence** and substituents are **cited alphabetically** in the name.



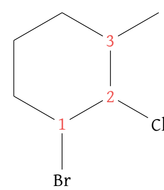
(a) 2-Chloro-1-iodo-4-methyl-cyclohexane



(b) 2-Chloro-3-ethyl-1,5-dimethyl-cyclooctane



(c) 1-Ethyl-2-methyl-cyclohexane



(d) 1-Bromo-2-chloro-3-methyl-cyclohexane

Figure 54: Cycloalkane nomenclature examples

(3) *Other rules follow the same process as those discussed in Core Concept 4.3*

4.2 Alkenes

Core Concept 4.6: Alkenes

Alkenes (*Section 1*) are unsaturated hydrocarbons containing at least one carbon–carbon double bond. Their nomenclature generally follows the same conventions as those of alkanes (*Core Concept 4.1*), except that the suffix ”-ane” is replaced with ”-ene” to denote the presence of a double bond (*e.g., ethene, propene, and butene*).

4.2.1 n-Alkenes

Core Concept 4.7: Alkenyl Substituents

An alkenyl **substituent** originates from an alkene through the removal of a hydrogen atom, while preserving the carbon–carbon double bond. These substituents generally follow the formula C_nH_{2n-1}, and are named by replacing the ”-ene” suffix of the parent alkene with ”-enyl” (*e.g., ethenyl, C_2H_3*). Additionally, the position of the double bond must be specified by indicating its **starting** location relative to the point of attachment to the parent chain, as illustrated by *prop-2-enyl*.

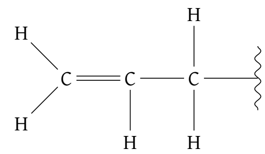


Figure 55: Prop-2-enyl substituent

When a substituent is **connected to the parent backbone through a double bond**, the suffix "-ylidene" is used in place of "-enyl" to indicate the mode of attachment. Observe that the position of the double bond does not require explicit specification, as it is inherently implied by the "-ylidene" suffix.

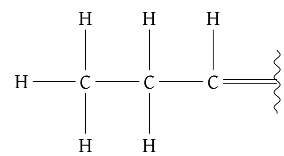
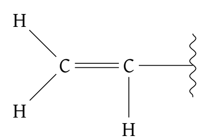
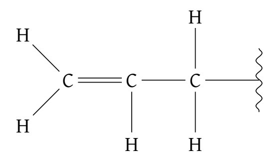


Figure 56: Propylidene substituent

Certain substituents possess historically established trivial names that remain prevalent in chemical nomenclature and must be committed to memory, as they do not follow systematic naming conventions.



Vinyl (*Ethenyl*)



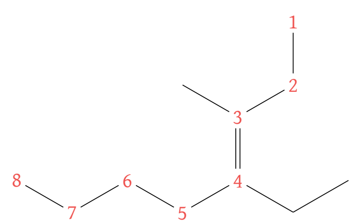
Allyl (*Prop-2-enyl*)

Figure 57: Common alkenyl substituent names

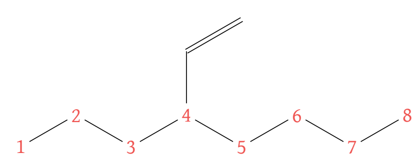
Core Concept 4.8: Alkene IUPAC System

The systematic nomenclature of alkenes follows the procedure outlined below.

(1) **Parent Chain Identification:** Identify the longest continuous carbon chain, irrespective of double bond incorporation. If both C_{sp^2} carbons are contained within this principal chain, the compound is designated as an alk-**ene**. Otherwise, it is named as an **alkenyl**-alkane.



4-Ethyl-3-methyl-oct-3-ene



4-vinyl-octane

Figure 58: Alkene nomenclature examples

(2) **Position Numbering**: Assign numerical locants such that the doubly bonded carbons receive the lowest possible numbers. When specifying the position of the double bond, only the lower-numbered of the two C_{sp^2} carbons is indicated.

(3) *Alkenes exhibit stereoisomerism arising from restricted rotation about the carbon-carbon double bond. Nomenclature conventions for disubstituted alkenes are detailed in Core Concept 4.24, while tri- and tetrasubstituted alkenes are addressed in Subsubsection 4.6.3.*

(4) *Other rules follow the same process as those discussed in Core Concept 4.3.*

A representative example demonstrating the application of these principles is provided below.

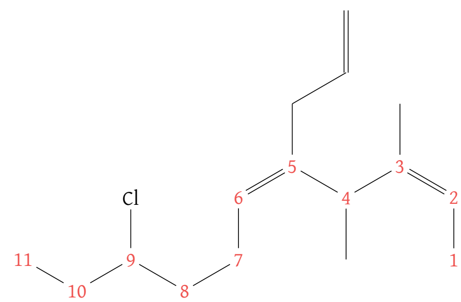
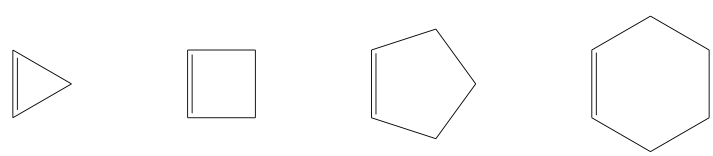


Figure 59: (2Z,5Z)-5-Allyl-9-chloro-3,4-dimethyl-undec-2,5-diene

4.2.2 Cycloalkenes

Core Concept 4.9: Cycloalkenes

Cycloalkenes are cyclic structures containing one or more carbon-carbon double bonds. Their systematic nomenclature is derived from the number of carbon atoms constituting the ring, with the prefix "cyclo-" appended to the corresponding alkene root name.



Cyclopropene Cyclobutene Cyclopentene Cyclohexene

Figure 60: Cycloalkenes

Core Concept 4.10: Cycloalkene IUPAC System

The systematic nomenclature of cycloalkenes adheres to conventions analogous to those governing cycloalkanes (*Core Concept 4.5*). A key distinction is that in cyclic systems, the doubly bonded carbons are assigned positions C^1 and C^2 by convention; consequently, explicit numerical designation of the double bond position is redundant. The assignment of subsequent positional locants follows the same lowest possible digit sequence rule established in Core Concept 4.5.

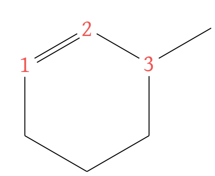


Figure 61: 3-Methyl-cyclohexene

4.3 Alkynes & Alkenynes

Core Concept 4.11: Alkynes & Alkenynes

Alkynes (*Section 2*) are unsaturated hydrocarbons containing at least one carbon–carbon triple bond. Systematic nomenclature is established by replacing the terminal "-ane" suffix of the corresponding saturated alkane (*Core Concept 4.1*) with "-yne" to denote the presence of triple-bond unsaturation (*e.g., ethyne, propyne, butyne*).

Alkenynes are hydrocarbons possessing both carbon–carbon double and triple bonds within the same molecular framework (*e.g., propenyne, butenyne, pentenyne*).

4.3.1 n-Alkynes

Core Concept 4.12: Alkynyl & Alkenynyl Substituents

An alkynyl **substituent** is formed from an alkyne by abstraction of a hydrogen atom, while maintaining the carbon–carbon triple bond. Their general formula is given as C_nH_{2n-3}, and their nomenclature is based on the parent alkyne name, with the suffix "-yne" changed to "-ynyl" (*e.g., ethynyl,* C_2H). Additionally, the position of the triple bond must be specified by indicating its **starting** location relative to the point of attachment to the parent chain, as illustrated by *prop-2-ynyl*. This convention extends analogously to alkenynyl substituents.

A directly analogous alkyne substituent corresponding to alkylidenes in alkenes is generally not encountered within the scope of these notes. This is because a branching carbon atom attached to the parent chain can maximally participate in a double bond, as its remaining valences are already occupied by bonds within the parent chain framework.

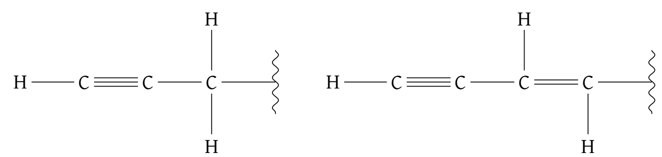


Figure 62: Prop-2-ynyl (*left*) and but-1-en-3-ynyl (*right*) substituent

Core Concept 4.13: Alkyne & Alkenyne IUPAC System

The systematic nomenclature of alkynes adheres to those established for alkenes (*Core Concept 4.8*). However, the nomenclature of alkenynes necessitates additional rules for positional numbering.

- Positional numbering commences from the terminus nearest to either unsaturated functional group, irrespective of bond multiplicity.

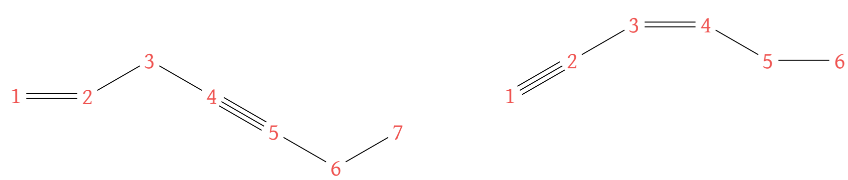


Figure 63: Hept-1-en-4-yne (*left*) and cis-Hex-3-en-1-yne (*right*)

- When both double and triple bonds are equidistant from the terminal positions, the double bond receives priority in numbering by virtue of alphabetical precedence (*"-ene-" precedes "-yne"*).

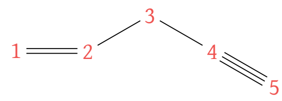


Figure 64: Pent-1-en-4-yne

4.4 Alcohol & Ethers

Core Concept 4.14: Alcohol & Ethers

Alcohols and **ethers** constitute a fundamental class of organic compounds characterized by the presence of one or more –OH and –OR functional groups, respectively, bonded to sp^3-hybridized carbon atoms.

4.4.1 n-Alcohol/Ethers

Core Concept 4.15: Alcohol IUPAC System

The systematic nomenclature of alcohols follows the procedure outlined below.

(1) **Parent Chain Identification**: Identify the longest continuous carbon chain **directly connected to the –OH group** and replace the suffix alkan-**e** with alkan-**ol**. When multiple –OH groups are present, the suffix -ol is modified to -diol, -triol, etc., accordingly. *Note that the selected chain may not necessarily be the longest carbon chain in the molecule.*

(2) **Alcohol Substituents**: When –ROH groups function as substituents rather than principal functional groups, they are designated as "hydroxyalkyl-" groups.

(3) **Position Numbering**: Initiate numbering from the terminus **nearest** to the –OH group. When met with multiple hydroxyl groups, continue comparison until reaching the **first point of difference** as shown in Figure 65.

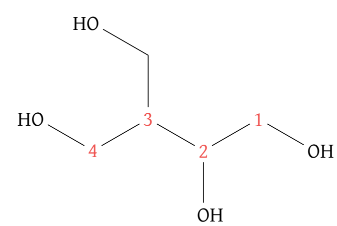


Figure 65: 3-Hydroxymethyl-butane-1,2,4-triol

(4) *Other rules follow the same process as those discussed in Core Concept 4.3.*

Representative examples illustrating the application of the aforementioned nomenclature protocol are provided below.

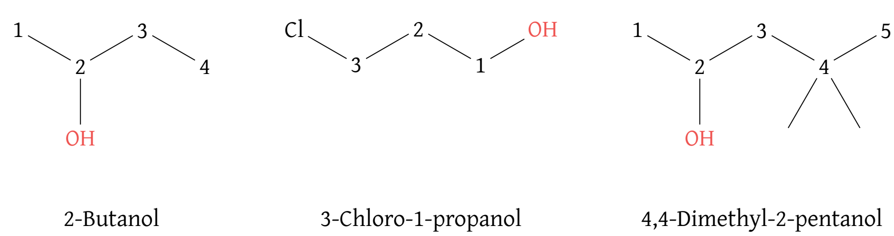
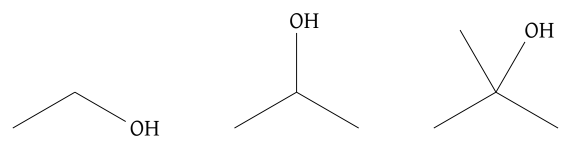


Figure 66: Alcohol nomenclature examples

Core Concept 4.16: Primary, Secondary, and Tertiary Alcohols

Analogous to the classification system employed for haloalkanes, alcohols are similarly categorized as primary, secondary, or tertiary based on the degree of substitution at the carbon atom bearing the hydroxyl group.

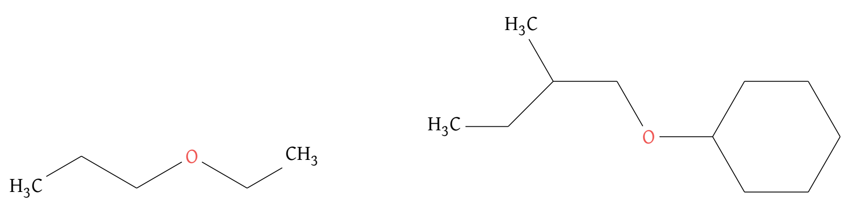


Primary alcohol Secondary alcohol Tertiary alcohol

Figure 67: Primary, secondary, and tertiary alcohols

Core Concept 4.17: Ether IUPAC System

In ether (R'OR) IUPAC nomenclature, the larger carbon-containing group is designated as the parent chain, while the smaller group attached to the oxygen is named as an "alkoxy-" substituent.



1-Ethoxypropane 2-Methylbutoxy-cyclohexane

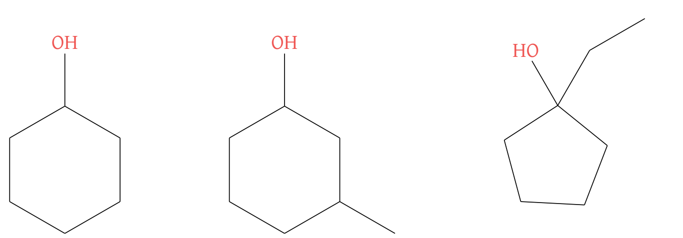
Figure 68: Ether nomenclature examples

Other rules follow the same process as those discussed in Core Concept 4.3.

4.4.2 Cycloalkanols

Core Concept 4.18: Cycloalkanols

Cyclic alcohols are designated **cycloalkanols** and adhere to conventions analogous to those of cycloalkanes (*Core Concept 4.5*). However, the carbon attached to the OH group is always designated as position 1.



cyclohexan-1-ol cis-3-methyl-cyclohexan-1-ol 1-ethyl-cyclopentan-1-ol

Figure 69: Cycloalkanol nomenclature examples

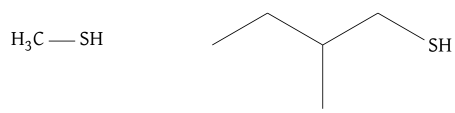
4.5 Thiols & Thioethers

Core Concept 4.19: Thiols & Thioethers

Thiols and **thioethers** (*Core Concept 3.1*) are classes of organic compounds characterized by the presence of sulfhydryl (–*SH*) and alkylthio (–*SR*) functional groups, respectively, bonded to a carbon atom. Structurally, thiols and thioethers are sulfur analogs of alcohols and ethers, respectively, in which the corresponding oxygen atom is replaced by sulfur.

Core Concept 4.20: Thiol IUPAC System

Thiols follow nomenclature conventions analogous to those of alcohols (*Core Concept 4.15*), with the primary distinction being the replacement of the suffix ”-ol” with ”-thiol”.



Methanethiol 2-Methyl-1-butanethiol

Figure 70: Thiol nomenclature examples

Similarly, when a sulfhydryl group functions as a **substituent**, it is designated by the prefix ”mercapto-” (–SH). This nomenclature is most commonly employed when both hydroxyl (–OH) and sulfhydryl (–SH) functional groups are present within the same molecule. In such cases, the hydroxyl group takes precedence in determining the principal functional group, and the compound is therefore classified and named as an alcohol rather than a thiol.

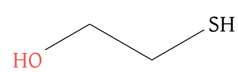


Figure 71: 2-Mercapto-1-ethanol

Core Concept 4.21: Thioether IUPAC System

Thioethers follow nomenclature conventions analogous to those of ethers (*Core Concept 4.17*), with the primary distinction being the use of the prefix ”alkylthio-” to designate –SR substituents.

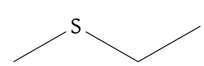


Figure 72: Methylthioethane

4.6 Isomer Configurations

Core Concept 4.22: Stereoisomers

Stereoisomers are defined as molecules possessing identical molecular formulas and constitutional connectivity of atoms, yet differing in the three-dimensional spatial arrangement of those atoms. Consequently, several distinct nomenclature systems have been developed to unambiguously distinguish among stereoisomers of a given molecular constitution.

4.6.1 Cis & Trans

Core Concept 4.23: Cycloalkanes

Cycloalkane rings possess two distinct spatial orientations relative to the plane of the ring: "above" and "below." When two or more substituents are present, those oriented on the same side of the ring are designated **cis**, while those positioned on opposite sides are designated **trans**.

cis-1-Chloro-2-methylcyclohexane trans-1-Chloro-2-methylcyclohexane

Figure 73: Cis and trans cycloalkane isomers

Substituents projecting away from the observer (*below the plane of the ring*) are represented by **dashed bonds**, whereas those projecting toward the observer (*above the plane of the ring*) are denoted by **wedged bonds**, as illustrated in Figure 73.

Core Concept 4.24: Alkenes

Disubstituted alkenes exhibit two distinct spatial orientations for their substituents. Analogous to the nomenclature conventions established for cycloalkanes (Core Concept 4.23), these geometric isomers are designated as **cis** or **trans** according to whether the substituents are oriented on the same or opposite sides of the double bond plane, respectively.

trans-pent-2-ene cis-pent-2-ene

Figure 74: Cis and trans alkene isomers

4.6.2 Rectus & Sinister (R/S)

Core Concept 4.25: Absolute Configuration

While the cis-trans nomenclature system is convenient and adequate for describing simple cycloalkane stereoisomers, more complex stereochemical situations necessitate an alternative system: **absolute configuration**.

The **absolute configuration** system provides a universal framework for unambiguously describing the three-dimensional arrangement of substituents at a stereogenic carbon atom in both acyclic and cyclic compounds.

Core Concept 4.26: Chiral Center

A **chiral center** (*or stereocenter*) is typically a carbon atom bearing four distinct substituents arranged in tetrahedral geometry.

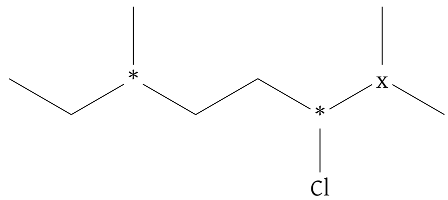


Figure 75: Chiral (*) and non-chiral (*x*) center examples

As illustrated above, each chiral center possesses four distinct substituents, which may be identified as follows.

- Left chiral center → ethyl, methyl, hydrogen, *rest of the molecule*
- Right chiral center → Chloro, isopropyl, hydrogen, *rest of the molecule*

Conversely, the non-chiral carbon is achiral due to the presence of two identical methyl substituents.

- Non-chiral center → methyl, methyl, hydrogen, *rest of the molecule*

Having identified the chiral center, the attached substituents must be ranked according to the **Cahn-Ingold-Prelog priority rules**, which are based on the atomic number of the atoms directly bonded to the stereocenter.

Consider the left chiral center, which is bonded to three carbon atoms and one hydrogen atom. Since hydrogen has a lower atomic number than carbon, it is immediately assigned the lowest priority (*rank 4*). When directly attached atoms are identical, priority is determined by proceeding outward along each substituent chain until the **first point of difference** is encountered.

- Ethyl → C, H, H
- Methyl → H, H, H
- Rest of the molecule → C, H, H

Based on this assessment of the atoms bonded to each substituent's first carbon, the methyl group is assigned priority rank 3. Continuing this analysis, the rest of the molecule (ROM) receives priority rank 1, and the ethyl group receives rank 2.

(1) ROM

(2) Ethyl

(3) Methyl

(4) Hydrogen

The designations **R** (*rectus*) and **S** (*sinister*) describe the rotational direction from priority 1 to 2 to 3: **clockwise** rotation corresponds to R, while **counterclockwise** rotation corresponds to S. This determination is made from a perspective in which the lowest-priority group (*rank 4*) is oriented directly away from the observer.

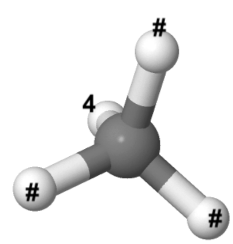


Figure 76: Absolute configuration perspective visualized

Core Concept 4.27: R/S Absolute Configuration Example (*Alkane*)

Since the stereochemistry (*wedge or dash configuration*) of the methyl group is not explicitly defined in Figure 75, the absolute configuration cannot be unambiguously assigned. Therefore, both possible stereochemical scenarios are examined below.

(a) **Methyl group is oriented as a wedge.**

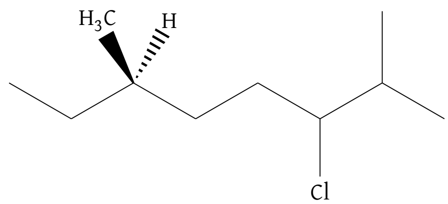


Figure 77: Left methyl substituent from Figure 75 as a wedge

In this orientation, the molecule is already positioned such that the lowest-priority group (*hydrogen, rank 4*) projects away from the observer. Tracing the path from rank 1 (*ROM*) through rank 2 (*ethyl*) to rank 3 (*methyl*) proceeds in a clockwise direction; consequently, this stereocenter is assigned the **R configuration**.

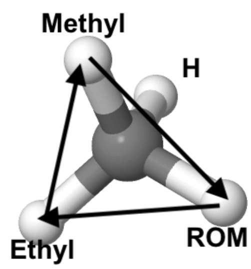


Figure 78: R configuration

(b) **Methyl group is oriented as a dash.**

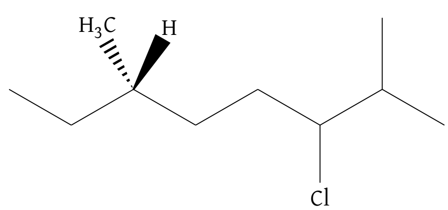


Figure 79: Left methyl substituent from Figure 75 as a dash

Application of the same analytical procedure reveals a counterclockwise progression from rank 1 through rank 2 to rank 3, thereby assigning this stereocenter the **S configuration**.

In systematic nomenclature, the R/S descriptor is indicated in parentheses at the beginning of the compound name, preceded by the numerical locant designating the position of the chiral center.

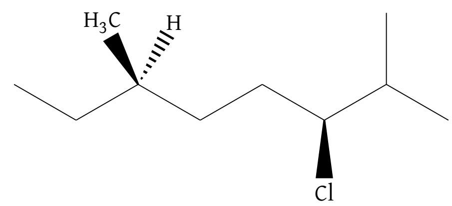


Figure 80: (3S,6R)-3-chloro-2,6-dimethyloctane

Core Concept 4.28: R/S Absolute Configuration Example (*Cycloalkane*)

The absolute configuration assignment procedure went over in Core Concept 4.27 is equally applicable to cycloalkane systems.

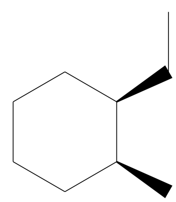
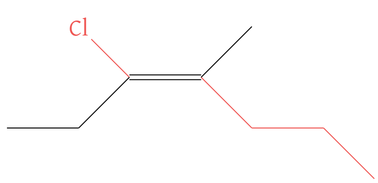


Figure 81: (1R,2S)-1-ethyl-2-methylcyclohexane

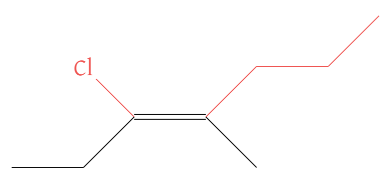
4.6.3 Entgegen & Zusammen (E/Z)

Core Concept 4.29: Entgegen & Zusammen (E/Z)

The E/Z nomenclature system applies the Cahn-Ingold-Prelog priority rules (*Core Concept 4.26*) independently to each C_{sp^2} carbon to identify the highest-priority substituent at each terminus of the double bond. When the respective higher-priority groups are positioned on opposite sides of the double bond plane, the configuration is designated as E, whereas placement on the same side denotes the Z configuration.



(3E)-3-Chloro-4-methyl-hept-3-ene



(3Z)-3-Chloro-4-methyl-hept-3-ene

Figure 82: E and Z configurations of 3-Chloro-4-methyl-hept-3-ene